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An improved artificial ecosystem optimization algorithm for optimal configuration of a hybrid PV/WT/FC energy system

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Abstract This paper mainly focuses on the optimal design of a grid-dependent and off-grid hybrid renewable energy system (RES). This system consists of Photovoltaic (PV), Wind Turbine (WT) as well as Fuel Cell (FC) with hydrogen gas tank for storing the energy in the chemical form. The optimal components sizes of the proposed hybrid generating system are achieved using a novel meta-heuristic optimization technique. This optimization technique, called Improved Artificial Ecosystem Optimization (IAEO), is proposed for enhancing the performance of the conventional Artificial Ecosystem Optimization (AEO) algorithm. The IAEO improves the convergence trends of the original AEO, gives the best minimum objective function, reaches the optimal solution after a few iterations numbers as well as reduces the falling into the local optima. The proposed IAEO algorithm for solving the multiobjective optimization problem of minimizing the Cost of Energy (COE), the reliability index presented by the Loss of Power Supply Probability (LPSP), and excess energy under the constraints are considered. The hybrid system is suggested to be located in Ataka region, Suez Gulf (latitude 30.0, longitude 32.5), Egypt, and the whole lifetime of the suggested case study is 25 years. To ensure the accurateness, stability, and robustness of the proposed optimization algorithm, it is examined on six different configurations, representing on-grid and off-grid hybrid RES. For all the studied cases the proposed IAEO algorithm outperforms the original AEO and generates the minimum value of the fitness function in less execution time. Furthermore, comprehensive statistical measurements are demonstrated to prove the effectiveness of the proposed algorithm. Also, the results obtained by the conventional AEO and IAEO are compared with those obtained by several well-known optimization algorithms, Particle Swarm Optimization (PSO), Salp Swarm Algorithm (SSA), and Grey Wolf Optimizer (GWO). Based on the obtained simulation results, the proposed IAEO has the best performance among other algorithms and it has

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successfully positioned itself as a competitor to novel algorithms for tackling the most complicated engineering problems.

1. Introduction

1.1. Motivations

Nowadays, the development of renewable energy resources technology goes very quickly. Despite this development, there is a big shortage of electrical energy in remote areas such as rural and island areas [1–7]. In recent years, the power electronic devices are widely developed and used to help in increasing the spread of these renewable sources of electrical energy [2,8–11]. Furthermore, the RES such as PV and WT, by their nature, have several fluctuations during their operation in the power system, and such fluctuations lead to instability in the power system operation [12]. These variations lead to an increase in the value of the produced voltage, thus, the required capacity of storage systems to store abundant energy from renewable sources will be increased [2,6]. Therefore, the combination of the different kinds of renewable sources of energy should be provided as well as adopting suitable systems of energy storage [5,13]. The generation mix of RES in one system is called a hybrid energy system. The remote areas mainly depend on diesel generators as their primary source of electrical energy.

Egypt is rich with many RES such as wind and PV sources, which are represented by the high solar radiation intensity and high continuous velocity of air that blows the wind turbines. In Egypt as one of the promising countries for implementing RES, the total installed capacity of New and renewable energy of PV and WT has been raised from 1157 MW in 2017/2018 to 2247 MW in 2018/2019 with an increasing rate of 94.2% as reported in the 2018/2019 annual report of the Egyptian Electricity Holding Company (EEHC) [14]. One should note that solar energy generation is increased by 184% from 537GWhGWh to 1525. The plane aims to increase the dependency on RES, for example, Oyoun Moussa thermal power plant fueled by coal will be replaced by a hybrid PV/WT system with an installed capacity of 750 MW [14]. Hence, there are several scientific pieces of research, and feasibility studies are conducted to determine the appropriate optimal places for the installation of new and renewable energy power plants. The proposed area for this study is called Ataka, which located in Suez city on the Red Sea shore (30.0° north latitude, 32.5° east longitude), Egypt. The solar irradiation and the air velocity in the proposed site are very high and suitable for the construction of different hybrid RESs. In this work, the proposed hybrid system is investigated under the grid-connected and the islanded cases to cover the load requirements in the proposed site. In 2015, Egypt installed a huge wind farm with a power capability of 380 MW in Gabal Elzeit. According to the reports of the ministry of electricity and energy in 2017 [14–16], Egypt plans to install new renewable power plants in the Suez Gulf region.

1.2. Literature review

Recently, several applications of modern optimization algorithms utilized for the optimal configuration of various hybrid RES have been provided in several scientific research papers. Ref. [17], applied two different optimization algorithms, namely, Genetic Algorithm (GA) and Particle Swarm Optimization (PSO) to obtain the optimal design of hybrid PV/WT/Biomass generating system through minimizing the objective function presented by the COE. Ref. [18], provided strength Pareto evolutionary method to determine the optimal components' sizes of hybrid PV/WT/diesel power system under multi-objective functions, minimizing COE and minimizing emissions of the greenhouse gases. Sajjad Abedi et al. [19], utilized Differential Evolutionary Algorithm (DEA) for the optimal configuration of grid-independent PV/WT/FC hybrid RES for minimizing the whole system cost along the overall lifetime of the proposed system. In [20], Monte Carlo simulations were utilized to evaluate the annual balance of energy and the network stress caused by power mismatch in RES. The evaluation of the overall performance of the hybrid system was based on the following criteria, reliability of annual energy balance, capital investment and grid.

In [21], the optimal configuration of standalone hybrid PV/WT/pumped storage system based on a techno-economic evaluation through lower COE and minimum LPSP as objective functions has been determined. M. A. Mohamed et al. [22], used the Cuckoo Search (CS) optimization algorithm for determining the optimal capacities of grid-independent hybrid PV/WT/ diesel/battery storage renewable generating systems. In [23], a stand-alone PV/WT/diesel engine hybrid renewable energy system has been presented and the optimal sizes of its components have been obtained using an adaptive differential evolution optimization algorithm. Ref. [2], applied three different optimization algorithms, namely, Water Cycle Algorithm (WCA), Moth-Flame Optimizer (MFO), Whale Optimization Algorithm (WOA), and Hybrid Particle Swarm-Gravitational Search Algorithm (PSOGSA) to generate the optimal component sizes of grid-independent PV/WT/diesel/battery storage subjected to minimum COE and LPSP.

Ref. [24] studied an off-grid PV/WT/battery/FC hybrid RES for the optimal design of its components using Improved Ant Colony Optimization (IACO) through the formulation of the following objective functions; minimum COE and maximum reliability of the proposed system. Refs. [25,26] used the PSO algorithm for the optimal configuration of PV/FC/WT off-grid hybrid RES. In [27], the optimal design of a stand-alone WT/Micro hydro/FC generating system has been achieved using GA.

Ref. [28], applied Biogeography Based Optimization (BBO) algorithm for determining the optimal configuration of PV/diesel generator/WT/battery storage off-grid hybrid energy system to deliver the electrical energy to the remote villages. A. Kamjoo et al. [29], used the Non-dominated Sorting

Genetic Algorithm (NSGA-II) to determine the optimal sizing of PV/WT/battery storage standalone hybrid power system. In [30], a Modified Firefly Algorithm (MFA) has been applied for generating the optimal components' capacities of a PV/WT/-battery storage grid-dependent system.

In [31], Improved Bee Algorithm (IBA) has been proposed and applied for optimal design of six different grid-independent hybrid renewable energy systems, namely, solar / battery / reverse osmosis desalination (ROD), wind/battery / ROD, solar / wind / battery / ROD, solar / hydrogen / ROD, wind / hydrogen / ROD, solar / wind / hydrogen / ROD. The main objective function has been adopted to minimize the cost of the total life cycle of the proposed hybrid system, which includes the area swept by the WT blades, the PV surface area, the number of hydrogen tanks, and the number of batteries. The simulation results obtained by IBA have been compared with the results of the simulation obtained by the Harmony Search Algorithm (HSA). Based on the simulation results, it was concluded that the proposed hybrid renewable energy system-based battery energy storage was more cost-effective than the hybrid system-based hydrogen energy storage system. Also, the results obtained by IBA were more reliable and precise than those obtained by HSA. In [32], the particle swarm optimization technique has been proposed to assess the influence of a grid-connected fuel cell-based combined heat and power (CHP) systems on the operational and performance cost of the proposed system subjected to the minimum cost of the proposed hybrid system.

In [33], Tabu Search (TS) optimization technique has been used for optimizing a small independent PV, WT, and battery hybrid power system. The main objective function has been adopted to minimize the life cycle cost of the proposed small independent hybrid system, which includes the batteries number, the PV surface area, and WT swept area. The simulation results obtained by TS have been compared with those obtained by the HSA-based load forecasting and simulated annealing algorithm-based load forecasting. The TS algorithm proved its reliability in solving the optimization problem compared with other optimization algorithms. In [34], a Modified Particle Swarm Optimization (MPSO) technique has been applied for minimizing the total cost of a grid-connected hybrid WT/FC and natural gas systems. It was found that the results obtained by MPSO were more reliable and accurate than those obtained by other optimization algorithms. W. Zhang et al. [35], used a hybrid optimization algorithm based on Chaotic Search, Harmony Search (HS), and Simulated Annealing (SA), for optimal sizing of a stand-alone hybrid PV/WT and hydrogen energy system. Minimizing the total life cycle cost has been considered as the main objective function of the proposed hybrid system and the LPSP has been taken into consideration to evaluate the reliability of the proposed system. A. Hatata et al. [36], utilized the Clonal Selection Algorithm to determine the optimum design of a PV/WT/Battery hybrid energy system by minimizing the whole cost of the proposed hybrid system. H. M. Sultan et al. [7], used a novel metaphor-less Rao optimization algorithm to solve the optimization problem of optimal design of a hybrid PV/WT/hydroelectric pumped storage (HPS) system. Minimizing the COE of the proposed hybrid system was the objective function. The optimization results obtained by Rao algorithm have been compared with those obtained by other well-known optimization algorithms. Rao algorithm proved its superiority in solving the proposed system compared to other algorithms. A. A. Zaki et al. [37], proposed a Modified Farmland Fertility Optimization Algorithm (MFFA) for optimal components design of a grid-connected hybrid PV, WT, and FC system through minimizing the COE. The results of the simulation obtained from the application of MFFA have been compared with the results obtained by the conventional Farmland Fertility Optimization Algorithm (FFA). A. A. Zaki et al. [1], proposed various metaheuristic optimization algorithms, namely, SSA, WCA, WOA, and GWO for optimal component sizes of hybrid PV/WT/hydroelectric pumped storage power system based on minimizing the COE of the proposed hybrid energy system. From this study, it has been concluded that the WOA was the best algorithm over the other algorithms. A Discrete Multiobjective Grey Wolf Optimization (DMGWO) algorithm has been developed and utilized for the optimal configuration of a hybrid RES while the capacity of the wind power plant is considered as a discrete variable applied by a corrective algorithm embedded in the proposed algorithm [38]. A. Fathy [39], applied Mine Blast Algorithm (MBA) to determine the optimal design of a hybrid PV/WT/ FC energy system for a remote area in Helwan city, Egypt based on minimizing the total annual cost of the hybrid system. The results obtained from the proposed MBA have been compared with those obtained by other well-known optimization algorithms, namely, PSO, Cuckoo Search (CS) algorithm, and Artificial Bee Colony (ABC) algorithm. From the obtained simulation results, it was concluded that the total annual cost of the proposed hybrid system using MBA was significantly reduced compared with PSO, ABC, and CSA. In [40], the optimal sizing for a grid-connected hybrid PV/wind/diesel/CHP generating system has been determined using MPSO algorithm. The main objective function has been adopted to minimize the total energy cost of the proposed hybrid system. The results obtained by the application of MPSO algorithm have been compared with those obtained by other algorithms. However, MPSO proved its high accurateness in solving the optimization problem of the proposed hybrid system. M. Samy et al. [41], used Firefly Algorithm (FA) for optimal configuration of a stand-alone hybrid PV/WT/FC energy system for a small-scale area in Egypt. Minimizing the total annual cost of the proposed hybrid system has been selected to be the main objective function. The optimization results obtained from the FA have been compared with those obtained by Shuffled Frog Leaping Algorithm (SFLA) and PSO.

Feasibility analysis of different configurations of hybrid microgrid systems based on renewable energy sources using the Hybrid Optimization Model for Electric Renewable (HOMER) software has been presented [42–46]. In [42], HOMER software has been used for optimal sizing of grid-independent PV/Desiel/battery storage hybrid RES in Malaysia where the obtained results proved that the proposed system is economically feasible, COE is nearing $0.895/kWh and NPC is $158,206. In [43], optimal sizing and sensitivity analysis of on-grid and off-grid hybrid PV/WT in Bangladesh has been presented. The sensitivity study showed that the obtained results are more sensitive to a small change in the input parameters such as the average solar radiation, average wind speed, the lifetime of the project, and the discount rate. In [44], the feasibility of hybrid RES in the southwest of Egypt using the real data of solar radiation and wind speed has been introduced. It has been concluded that a hybrid PV/WT/Desiel microgrid with a battery storage system ensures the minimum cost of energy generated from the hybrid system. Similarly, the feasibility of the same configuration has been validated using HOMER for a case study in Indonesia [45] and the results showed that the NPC of the optimized system reduced by 29.65% while the CO2 emissions reduced by 16 tons per year. In [46], PV uncertainties have been modeled using a Markovian process and other components are modeled as Markov processes with states, which depend on the states of the PV system. Finally, the combinatorial optimization problem was solved using branch-and-cut.

1.3. Contribution and paper organization

The main goal of this study is to obtain the optimal sizes of the components of different configurations of grid-dependent and isolated PV/WT/FC generating systems. To achieve this goal, both proposed IAEO and original AEO techniques are applied. The main objective function in all configurations of systems is to minimize the COE produced from the proposed generating system as well as ensuring the high reliability of the power supply. The main contributions of the provided research work could be summarized as follows:

- Proposing the IAEO technique to modify the performance of AEO as a recent metaheuristic optimization algorithm.
- The proposed IAEO, as well as conventional AEO techniques, are applied for determining the optimal capacities of different configurations of proposed systems.
- The viability and accuracy of the proposed IAEO are tested using three different configurations of the grid-connected hybrid system and other three isolated systems.
- The obtained results from the proposed IAEO are comprehensively compared with those obtained by the conventional AEO.
- Statistical measures are conducted to confirm the accuracy, stability, and robustness of the proposed IAEO in solving the studied optimization problem.
- The results confirm the robustness and stability of the proposed algorithm in generating the optimal design of different configurations of the proposed hybrid RES.

This paper is formulated in the following manner. Section II provides the subsystems' mathematical model of the proposed hybrid power systems. Section III introduces briefly the operation strategy of the hybrid systems. Section IV presents the optimization problem and objective functions. Section V introduces in detail the procedures of the proposed optimization algorithm. The case study is presented in Section VI. Sections VII discusses the results of the simulation, and finally, the extracted conclusions are summarized in Section VIII.

2. Mathematical description of proposed hybrid system components

The schematic diagram of the suggested grid-connected PV/WT/FC hybrid RES is shown in Fig. 1. The hybrid system includes PV energy plant, WT energy plant, FCs, inverter device, a tank of hydrogen, and water electrolyzer. This system will be investigated under stand-alone and grid-connected configurations.

The main strategy steps for the operation of the proposed system could be explained as follows:

- The extra electrical energy over the load needs, that supplied by PV and WT subsystems is fed to the water electrolyzer, which helps to store it as chemical energy through splitting the water components to hydrogen and oxygen gases.
- The extra electrical energy over the water electrolyzer capacity is transferred to the national grid.
- The shortage in the renewable electrical energy generated from PV and WT systems to feed the load is covered by the chemical energy of the stored hydrogen in the special tanks through the FC system.
- When the level of hydrogen in the tank becomes below the lowest allowable level, the shortage in the electrical energy required to store the hydrogen gas in the tank is dispatched from the national grid.

The mathematical models of all components included in the proposed hybrid system are briefly presented in the next parts.

2.1. PV model

The electrical power generated from series-connected several PV cells is expressed as follows [47–49]:

$$P_{PV}(t) = N_{PV} \times A_{PV} \times \eta_{PV} \times G(t) \tag{1}$$

where, N_{PV} denotes the number of photovoltaic modules, A_{PV} is the area of the PV module, η_{PV} represents the PV system efficiency, $G(t)$ refers to the ambient intensity of solar irradiation. The efficiency of the PV system can be calculated as follows [50]:

$$\eta_{PV}(t) = \eta_r \times \eta_t \times \left[1 - \beta_T \times (T_c(t) - 25) - \beta_T \times G(t) \times \left(\frac{T_{nom} - 20}{800}\right) \times (1 - \eta_r \times \eta_t)\right] \tag{2}$$

where, η_r and η_t denote the reference efficiency and the efficiency of the maximum power point tracking (MPPT) system, respectively. β_T represents the coefficient of temperature. T_C and T_{nom} present the current cell temperature, and the nominal operating cell temperature, respectively.

2.2. B. Wind turbine model

The fundamentals of aerodynamics state that the speed of wind varies with height. The speed of wind at a specific height like WT hub, in terms of the speed of the wind, acts at the point of measurement can be obtained from the next expression [47]:

$$\frac{v_2}{v_1} = \left(\frac{h_2}{h_1}\right)^{\alpha} \tag{3}$$

where, v_2 denotes the speed of wind at WT hub at a certain height (h_2), and v_1 represents the speed of wind at the anemometer height h_1, α is the friction coefficient. The power of the wind turbine at a specific wind speed v can be formulated as follows [47,51]:

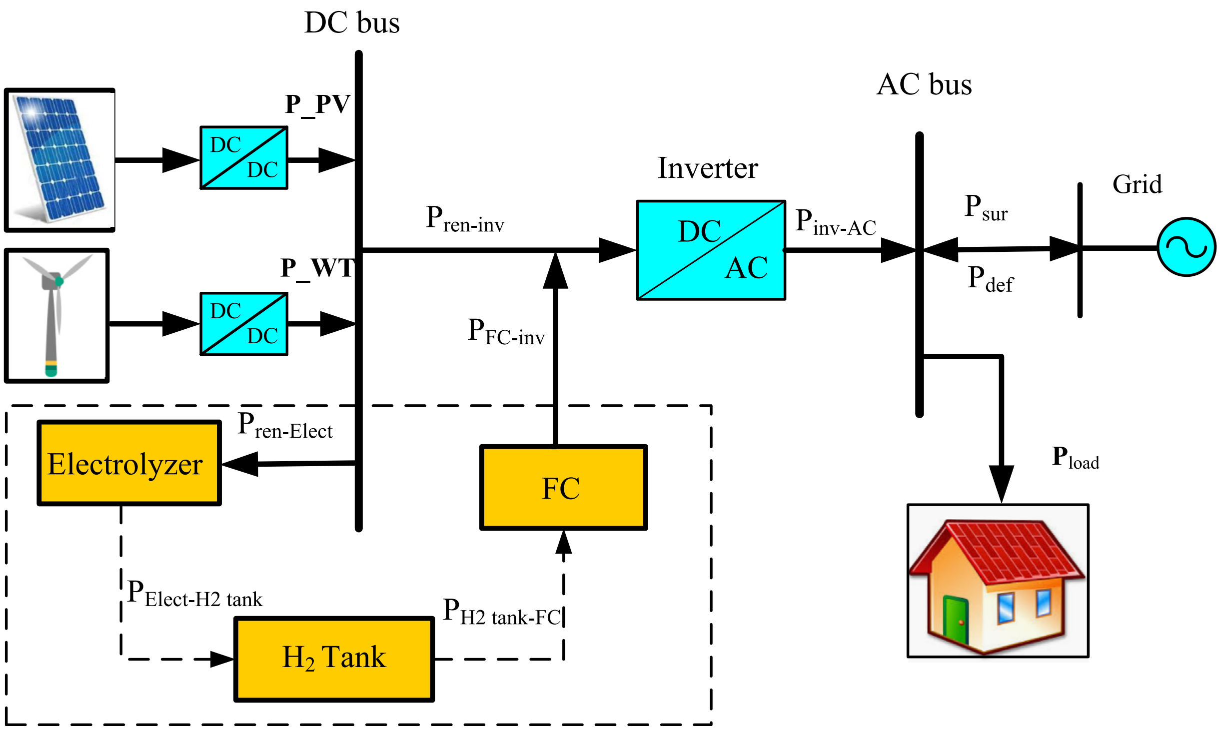

Fig. 1 Schematic configuration of the proposed grid-connected PV/WT/FC hybrid RES.

$$P_{WT}(t) = \begin{cases} 0 & v(t) < v_{cut-in} \\ a \times v(t)^3 - b \times P_{WT\_rated} & v_{cut-in} < v(t) < v_{rated} \\ P_{WT\_rated} & v_{rated} < v(t) < v_{cut-off} \\ 0 & v(t) > v_{cut-off} \end{cases} \tag{4}$$

$$\begin{cases} a = \frac{P_{WT\_rated}}{(v_{rated})^3 - (v_{cut-in})^3} \\ b = \frac{v_{cut-in}}{(v_{rated})^3 - (v_{cut-in})^3} \end{cases} \tag{5}$$

where, P_{WT_rated} is the rated electrical power of WT at rated speed (v_{rated}); $v_{cut\text{-}in}$ is the speed of wind at which the WT starts generating electrical energy. $v_{cut\text{-}off}$ is the speed of wind at which WT starts turning off to be in the safe mode.

The rated power of the wind turbine can be expressed as follows:

$$P_{WT\_rated} = \frac{1}{2} \times \rho \times A_{WT} \times C_p \times (v_{rated})^3 \tag{6}$$

where, ρ denotes the density of air, A_{WT} is the swept area bt the blades of the WT, C_P is the power coefficient which ranged between 0.25 and 0.45. The proposed type of WT in this study is BWC Excel-R/48 from Bergey Wind Power characterized by 7.5 kW and 48 V DC. The main specifications of WT contain the cost and efficiency of the DC/DC converter.

2.3. Water electrolyzer, hydrogen tank, and FC models

The electrolyzer is used to analyze water molecules into its main component (hydrogen and oxygen) through flowing the direct current from the electrode side to another. The generated hydrogen is collected around the surface of the positive electrode (anode). According to [52] in most electrolyzers, the hydrogen is collected at a pressure of 30 bar, but the hydrogen is supplied to the FC at a low pressure of 1.2 bar. There are two ways for storing the produced hydrogen; in most studies, the produced hydrogen is directly transferred to the hydrogen tank, while in other studies the pressure is raised to 200 bar with the help of a compressor and the reason for that is to raise the stored energy in the tank [52]. There is a third opinion to use two tanks, low-pressure tank to which the hydrogen is transferred at low pressure. If this tank is fully charged a compressor is used to handle the hydrogen to the high-pressure tank to reduce the energy consumed by the compressor as possible as it will not be in operation all time [53]. In the present study, the collected hydrogen is directly transferred to the hydrogen tank at a pressure of 30 bar. Water electrolyzer model can be easily expressed as follows [47,54]:

$$P_{Elect-\text{tank}} = P_{ren-Elect} \times \eta_{Elect} \tag{7}$$

where $P_{ren\text{-}Elect}$ is the electrical power resulting from the PV and WT sources towards the water electrolyzer, η_{Elect} indicates to the water electrolyzer efficiency, and $P_{Elect\text{-}tank}$ represents power supplied to the hydrogen tank from the water electrolyzer. The efficiency is considered a constant value during the all-time of the simulation process.

The hydrogen tank model emphasizes the chemical energy stored in the tank at any time interval during system operation. This electrical energy stored inside the tank E_{tank}(t) can be expressed as follows [54,55]:

$$E_{\text{tank}}(t) = E_{\text{tank}}(t-1) + \left(P_{Elect-\text{tank}}(t) - \frac{P_{\text{tank}-FC}(t)}{\eta_{storage}}\right) \times \Delta t \tag{8}$$

where E_{tank}(t-1) represents the electrical energy stored inside the hydrogen gas tank in the previous interval, $P_{tank\text{-}FC}$ denotes the power flow from hydrogen tank towards the

PEMFC, $\eta_{storage}$ is the efficiency of hydrogen gas tank, which roughly equals 95% and Δt represents the time intervals between simulating points, which is considered as one hour in this work.

The electrical energy accumulated in the tank of hydrogen gas is computed as a function of the hydrogen mass as follows [47]:

$$M_{tank}(t) = \frac{E_{tank}(t)}{HHV_{H_2}} \tag{9}$$

where, HHV_{H2} is the hydrogen gas higher heating value, which is usually taken as 39.7 kWh/m^2 [56]. The mass of stored hydrogen at any time during the operation of the proposed system should be between the allowable maximum ($M_{tank,\ max}$) and minimum ($M_{tank,\ min}$) bounds as follows:

$$M_{tank,min} \leqslant M_{tank}(t) \leqslant M_{tank,max} \tag{10}$$

PEMFC energy conversion efficiency is formulated as follows:

$$P_{FC-inv} = P_{tank-FC} \times \eta_{FC} \tag{11}$$

where η_{FC} is the electrical efficiency of PEMFC and it is assumed as 50% in this work [37,52].

2.4. Inverter model

The DC/AC inverter electronic device is utilized to provide the alternating current voltage to consumers. The inverter model is expressed in terms of the output power delivered to the load as follows:

$$P_{inv-AC} = (P_{FC-inv} + P_{ren-inv}) \times \eta_{inv} \tag{12}$$

where $P_{ren\text{-}inv}$ is the electrical power provided to consumers from the RES (PV and WT) and η_{inv} represents the inverter efficiency and it is assumed to be 90% in this work [54].

3. Operating strategy of the proposed hybrid system

To avoid repetition and length of the paper, the operation of one configuration is only introduced in this section. The operation of the proposed grid-dependent hybrid PV/WT/FC power system is briefly introduced based on the following operating scenarios:

- When the power supplied by the photovoltaic and wind resources, $P_{ren}(t) = P_{PV}(t) + P_{WT}(t)$, equals the demand load, $P_{load}(t)$, the electrical load is covered by the renewable generation without the need of any power from the FC system, $P_{ren}(t) = P_{load}(t)/\eta_{inv}$.
- When the electrical energy supplied by the RES is higher than the load, $P_{ren}(t) > P_{load}(t)/\eta_{inv}$, the difference is transferred to the water electrolyzer to generate hydrogen. In that regard, the surplus electrical energy is fed to the external electric network when the difference is bigger than the capacity of the electrolyzer.

$$P_{sur}(t) = P_{ren-inv}(t) \times \eta_{inv} - P_{load}(t) \tag{13}$$

- When the load power is higher than the renewable generated power, $P_{ren}(t) < P_{load}(t)/\eta_{inv}$, the difference in the consumption is supplied by the FC system. When the difference between the generation and load is bigger than the output power from the FC, the shortage in power can be taken from the grid utility.

$$P_{def}(t) = P_{load}(t) - (P_{ren-inv}(t) + P_{FC-inv}(t)) \times \eta_{inv} \tag{14}$$

The flowchart in Fig. 2 summarizes the strategy followed in the operation of the proposed hybrid PV/WT/FC energy system.

4. Formulation of the optimization problem

4.1. Cost of energy

The main goal of the optimization problem in this paper is to supply the load with electric energy from the proposed renewable hybrid system at minimum COE while ensuring the high reliability of the power supply. The COE is a function of the net present cost (NPC), which is an indication of the system annual cost. The annual cost of the proposed hybrid system (C_{ann_tot}) comprises the annual interest of capital cost, the operation and maintenance cost which is the lowest part in the case of renewable systems, the cost of replacements, and the penalty cost. For a grid-connected system, two components should be added to the annual cost; the cost of energy purchased from the external grid and the cost of energy sold. The NPC of the studied system is formulated as follows [22]:

For each subsystem i, the annual interest of capital cost is calculated according to the following [22]:

$$C_{ann\_cap\_i} = C_{cap\_i} \times CRF(r, M_i) \tag{15}$$

where C_{cap_i} denotes the capital investment cost of the subsystem, r is the rate of interest (here = 0.06), and M_i is the lifespan for each component. CRF for each subsystem is expressed as follows [22]:

$$CRF(r, M_i) = \frac{r(1+r)^{M_i}}{(1+r)^{M_i} - 1} \tag{16}$$

The annual cost for the replacement (C_{rep_ann}) for each subsystem i is calculated as follows [57]:

$$C_{ann\_rep} = C_{rep\_i} \frac{(M_{sys} - M_i)}{M_i} \tag{17}$$

where M_{sys} is the lifetime of the whole hybrid system (here 25 years). When the lifetime of the subsystem is less than the lifetime of the project, it means the replacement cost will be included.

When power supplied by the renewable sources exceeds the load consumption plus the power consumed by the electrolyzer, the surplus directed to the external grid (C_{gs}). Oppositely, if the PV, WT, and FC failed to satisfy the load demand, the deficit is supplied by the external grid (C_{gp}). The cost of energy sold and purchased are given as:

$$C_{gp} = N_{gp} \times C_p \tag{18}$$

$$C_{gs} = N_{gs} \times C_s \tag{19}$$

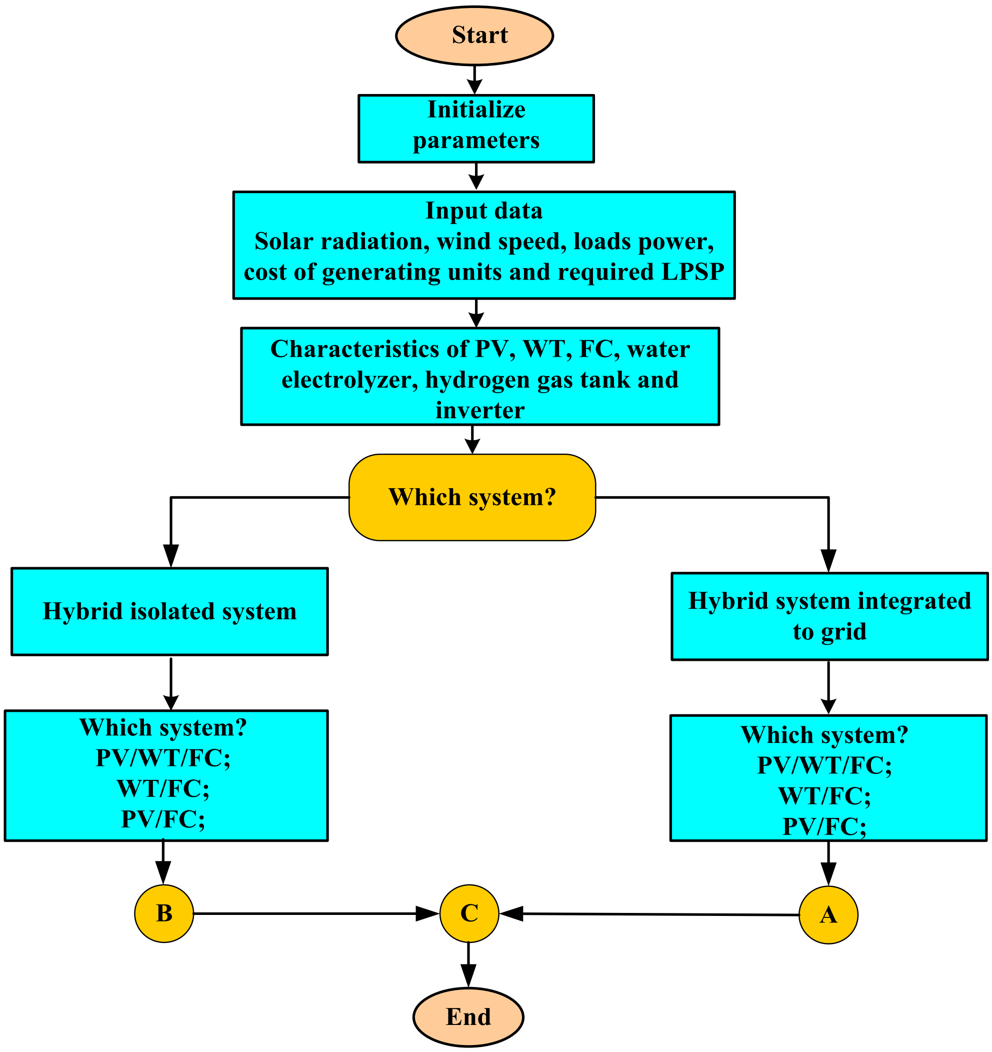

Fig. 2 Flowchart summarizing the operation of the proposed PV/wind/FC hybrid energy system.

where C_p (here 0.08\$/kWh) represents the cost of kWh taken from the grid, C_s (here 0.2\$/kWh) is the cost of kWh sold to the grid, N_{gp} is the total kWh units taken from the grid, and N_{gs} is total kWh units given to the grid.

$$NPC = \frac{C_{ann\_tot}}{CRF} \tag{20}$$

where CRF is the capital recovery factor of the system. The COE in (\$/kWh) supplied by the hybrid system is calculated as follows [57,58]:

$$COE = \frac{C_{ann\_tot}}{\sum_{h=1}^{h=8760} P_{load}} = \frac{NPC}{\sum_{h=1}^{h=8760} P_{load}} \times CRF \tag{21}$$

If the proposed hybrid system failed to satisfy the load demand at the predefined reliability or the fluctuation level of the power supplied to the external grid exceeds the allowable limit, an additional cost will be added to the whole cost of the system, called penalty cost. The penalty cost (C_{pc}) in the proposed system is an indication of the shortage in achieving the reliability index of the system, which is defined by the LPSP and the power fluctuation rate (D_{gs}). According to [57], the LPSP is limited by the predefined value (β_L) which is taken as 0.05. The fluctuation rate of the surplus power (D_{gs}) is controlled by the predefined value β_g, which should not higher than 33% of the installed capacity of the system within ten minutes as an interval of time.

$$C_{pc} = C_{pc\_1} \times (LPSP - \beta_L) \times \sum_{i=1}^{N} P_{load}(t_i) + C_{pc\_2} \times \frac{D_{gs} - \beta_g}{\beta_g} \times 100 \tag{22}$$

$$LPSP = \frac{\sum_{i=1}^{8760} [P_{load}(t_i) - (P_{PV}(t_i) + P_{WT}(t_i) + P_{FC-inv}(t_i))]}{\sum_{i=1}^{8760} P_{load}(t_i)} \tag{23}$$

$$D_{gs} = \frac{P_{sur\_\max} - P_{sur\_\min}}{\Delta t} \tag{24}$$

where C_{pc_1} is the cost applied when the predefined LPSP is not satisfied (here C_{pc_1} = 100\$/kWh) and C_{pc_2} is the cost included if the fluctuation exceeds the predefined level (here C_{pc_2} = \$50000/%). $P_{sur\_\max}$ and $P_{sur\_\min}$ are

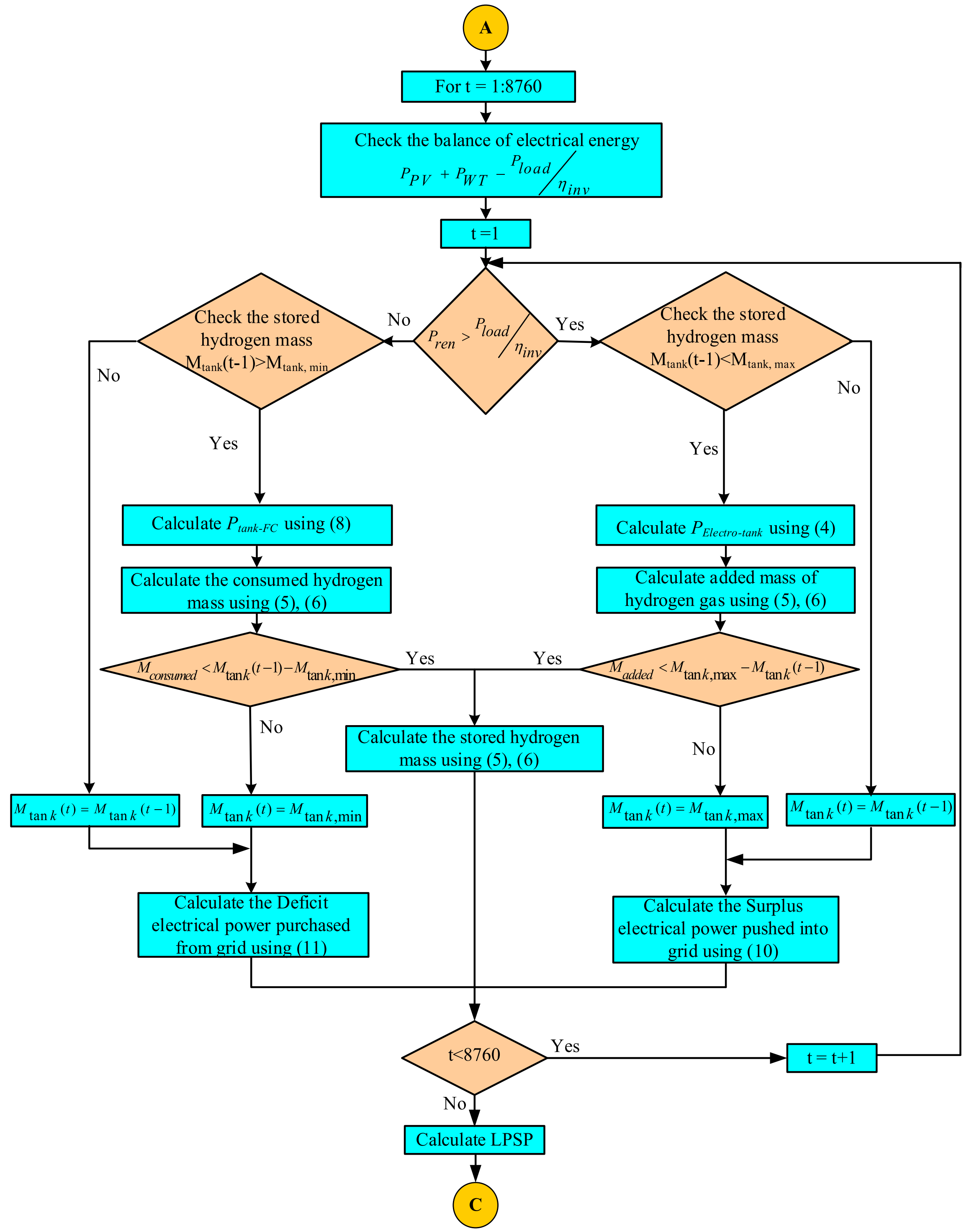

Fig. 2 (*continued*)

the maximum and minimum power injected into the external grid.

In this paper, a relative fluctuation rate (D_{load}) has been proposed to ensure that the renewable sources provide the most of the load demand and the load is mainly operated on the power supplied by the PV and wind systems. According to [57], the relative fluctuation rate is calculated as:

$$D_{load} = \frac{\sqrt{\frac{1}{N}\sum_{i=1}^{N}\left(P_{PV}(t_i) + P_{WT}(t_i) - P_{load}(t_i)\right)^2}}{\bar{P}_{load}} \quad (25)$$

where, $\bar{P}_{load}$is the average value of the load demand. According to [57], when D_{load} is small, it means better utilization of the RES and the generation is closer to the load consumption. The examined system indicates a high utilization of PV and wind resources if D_{load} is smaller than the allowable reference

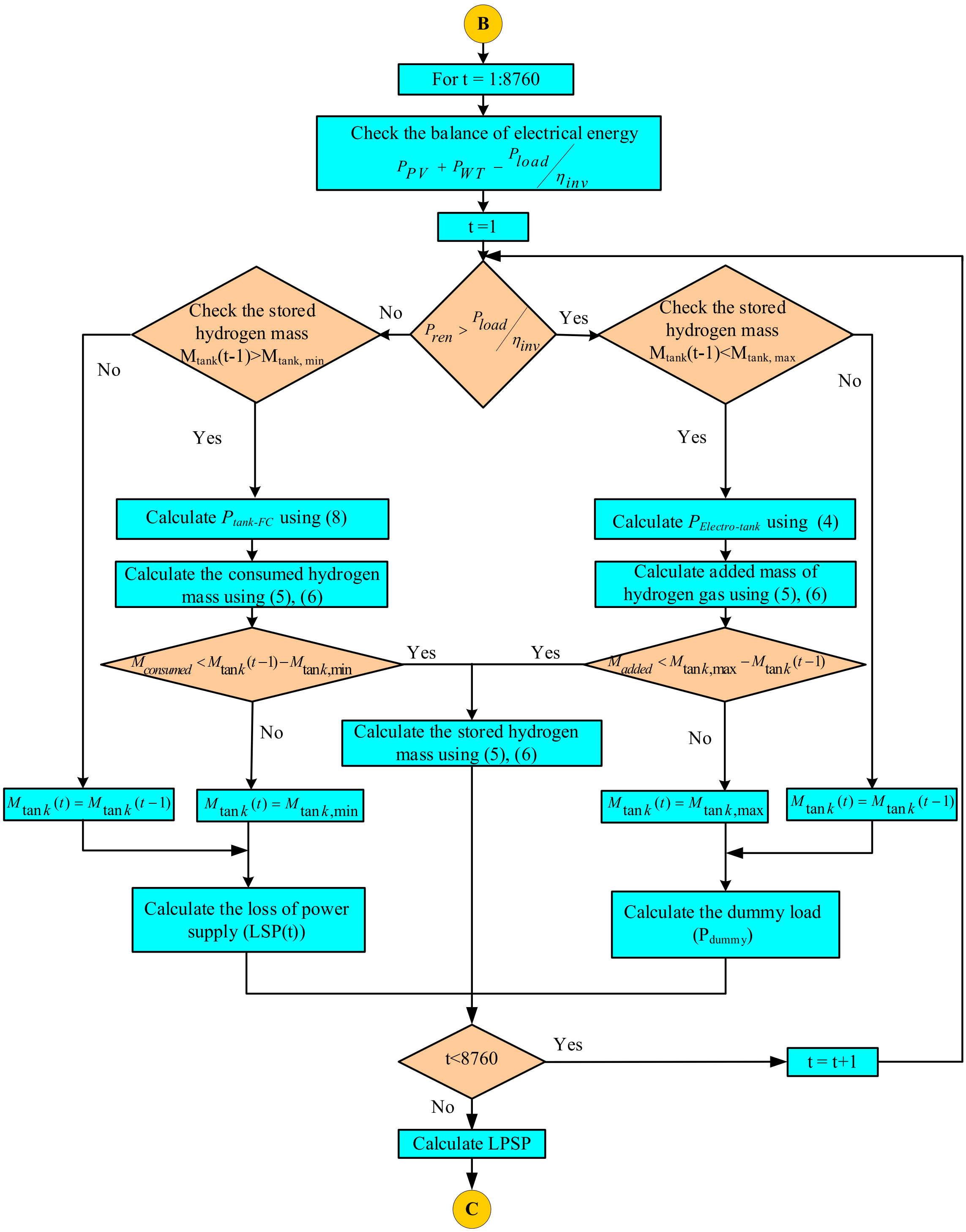

Fig. 2 (*continued*)

($\varepsilon_L = 2$). The specifications of all components included in the system are summarized in Table 1.

4.2. Objective function

In this research paper, two different objective functions are utilized. The three grid-connected configurations, PV/FC, WT/FC, and PV/WT/FC are optimally designed according to one objective function. For the grid-connected systems, the objective function is given in (26), which ensures the minimization of COE and LPSP as well as minimization of the relative power fluctuation rate (D_{load}) and fluctuation rate of power injected to the grid (D_{gs}).

$$\min f = \min(\lambda_1 \times COE + \lambda_2 \times LPSP + \lambda_3 \times D_{gs} + \lambda_4 \times D_{load}) \tag{26}$$

Table 1 Specification of all components included in the studied system [54]

Component	WT	PV modules	Electrolyzer	Hydrogen tank	FC	Inverter
Capital cost (US$/unit)	19,400	7000	2000	1300	3000	800
Replacement cost (US$/unit)	15,000	6000	1500	1200	2500	750
O&M cost (US$/ unit-year)	75	20	25	15	175	8
Lifespan (year)	20	20	20	20	5	15
Efficiency (%)	–	–	75	95	50	90
Power	7.5 kW	1 kW	1 kW	1 kW	1 kW	1 kW

where λ_1, λ_2, λ_3, and λ_4 are determined based on trial and error principle till reaching the optimal values. The IAEO and conventional AEO algorithms are applied for generating the optimal design of the proposed configurations using the following constraints:

$$\begin{aligned} & LPSP \leqslant \beta_L \\ & M_{\mathrm{tank,min}} \leqslant M_{\mathrm{tank}}(t) \leqslant M_{\mathrm{tank,max}} \\ & P_{dummy}(t) \leqslant P_{dummy,Max} \\ & P_{dummy}(t) \leqslant P_{dummy,Max} \end{aligned} \tag{27}$$

In the case of the isolated configuration, when the electrical power produced from PV and WT resources is higher than the load needs plus the rated power of the electrolyzer, a dummy load is used for generation-demand balance. The power dissipated in this load has to be minimized in order to minimize the total system cost. The total power of the dummy load relative to the total annual load demand (P_{dum}) is added in the objective function related to the standalone configurations as described in (28) and the corresponding constraints are summarized in (29):

$$\min f = \min(\lambda_1 \times COE + \lambda_2 \times LPSP + \lambda_3 \times P_{dum}) \tag{28}$$

$$\begin{aligned} & LPSP \leqslant \beta_L \\ & M_{\mathrm{tank,min}} \leqslant M_{\mathrm{tank}}(t) \leqslant M_{\mathrm{tank,max}} \\ & P_{dummy}(t) \leqslant P_{dummy,Max} \\ & P_{FC-inv}(t) \leqslant P_{FC,Max} \end{aligned} \tag{29}$$

5. Optimization algorithms

This section presents the conventional AEO and proposed IAEO techniques.

5.1. AEO

AEO is a nature-inspired optimization algorithm, inspired by the flow of energy in an ecosystem on the earth [59]. The ecosystem can be expressed as a group of living organisms live in a certain space and the ecosystem describes the relations between them [60]. The AEO mimics three unique behaviors of living organisms, which is including three phases, production, consumption, and decomposition. In the first phase (producers), it does not need to obtain energy from other organisms, where most producers are any kind of green plant. In the second phase (consumers), animals are represented as consumers, where they cannot make their food. These animals obtain nutrients from a producer or other consumers. The last phase is decomposers, where feeds on both producers (dead plants) or consumers (waste from living organisms). The three phases interacting with one another compose a food chain.

The AEO algorithm utilizes these three phases. The first phase is mainly to improve the balance between exploitation and exploration. The second phase is used to improve the exploration of the algorithm. In the last phase, it is proposed to enhance the exploitation of the AEO algorithm [59]. In the AEO algorithm, there is only one decomposer and one producer. The other individuals are considered consumers.

5.1.1. Producer

In this phase, a new individual is randomly produced between an individual randomly initiated in the search space (x_{rand}) and the best individual (x_n). The worst individual (producer) is updated by the upper and low boundaries of search space and the best individual (decomposer). This updated individual guides other individuals to search for different regions. This behavior participates in the balance between the explorative and exploitative search. The mathematical model for this phase can be expressed as follows [59]:

$$x_1(t+1) = (1-a)x_m(t) + a x_{rand}(t) \tag{30}$$

$$a = (1 - tMax.Iter)k_1 \tag{31}$$

$$x_{rand} = k(UP - LW) + LW \tag{32}$$

where, k_1 is a random number between [0, 1], $Max.Iter$ is the maximum number of iterations, a is a coefficient used for linear weighting, m is the number of populations, UP and LW are the lower and upper boundaries, respectively, and k is a random vector between [0, 1].

5.1.2. Consumption

The consumption allows the algorithm to update the solutions of individuals concerning either the solution offered by the solution of the randomly chosen individual with higher energy level, the production process or, both. It can enhance the exploration of the algorithm [59].

Levy flight is used to enhance the exploration level and mimics the food searching mechanism of many animals. The Levy flight can be described as follows [59]:

$$C = \frac{1}{2}\frac{v_1}{|v_2|} \tag{33}$$

$$v_1 \tilde{\mathrm{N}}(0,1),\ v_2 \tilde{\mathrm{N}}(0,1) \tag{34}$$

where N (0,1) represents a normal distribution.

This consumption factor will help each of the consumers to get food using different hunting strategies [59]. There are three types of consumers as:

- Herbivore type, that eats only producers and consumers.
- Carnivore type, that eats only the consumers with a higher energy level.
- Omnivore type, that eats other consumers with higher energy levels and/or producers.

Based on a random selection, the consumer can be classified as one of the three mentioned types. The mathematical model for the consumer, when it is selected as a herbivore type, can be represented as follows:

$$x_i(t+1) = x_i(t) + C.(x_i(t) - x_1(t)), \ forall \quad i \in (2, \dots, n) \tag{35}$$

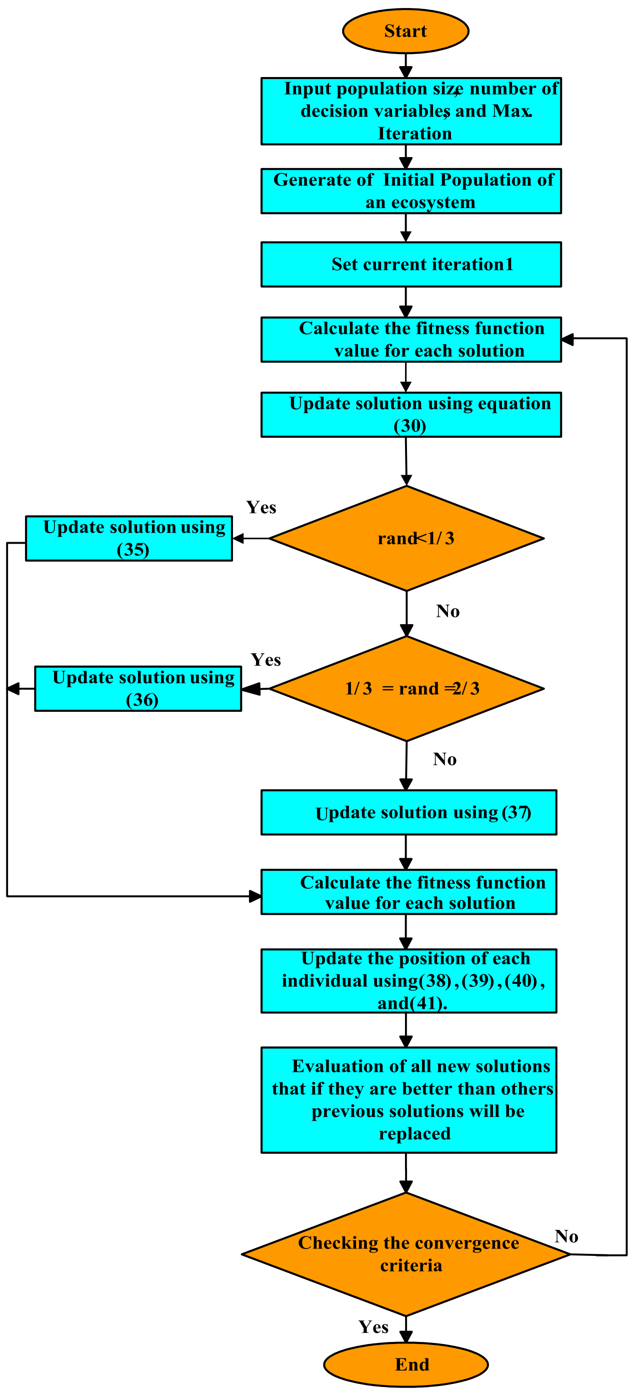

Fig. 3 Flowchart of the conventional AEO algorithm.

The mathematical model for the consumer, when it is selected as carnivore type, can be expressed as follows:

$$\begin{cases} x_i(t+1) = x_i(t) + C.(x_i(t) - x_j(t)), \ forall & i \in (3, \dots, n) \\ suchthat & j = randi([2i-1]) \end{cases} \tag{36}$$

The mathematical model for the consumer, when it is selected as omnivore type, can be expressed as follows:

$$\begin{cases} x_i(t+1) = x_i(t) + C.(r_2(x_i(t) - x_j(t)) \\ +(1-r_2)(x_i(t) - x_j(t))), & i = 3, \dots, n \\ suchthat & j = randi([2i-1]) \end{cases} \tag{37}$$

5.1.3. Decomposition

The decomposition enables AEO to update the solutions of individuals based on the best solution in the population via three decomposition coefficients, which include factor D and

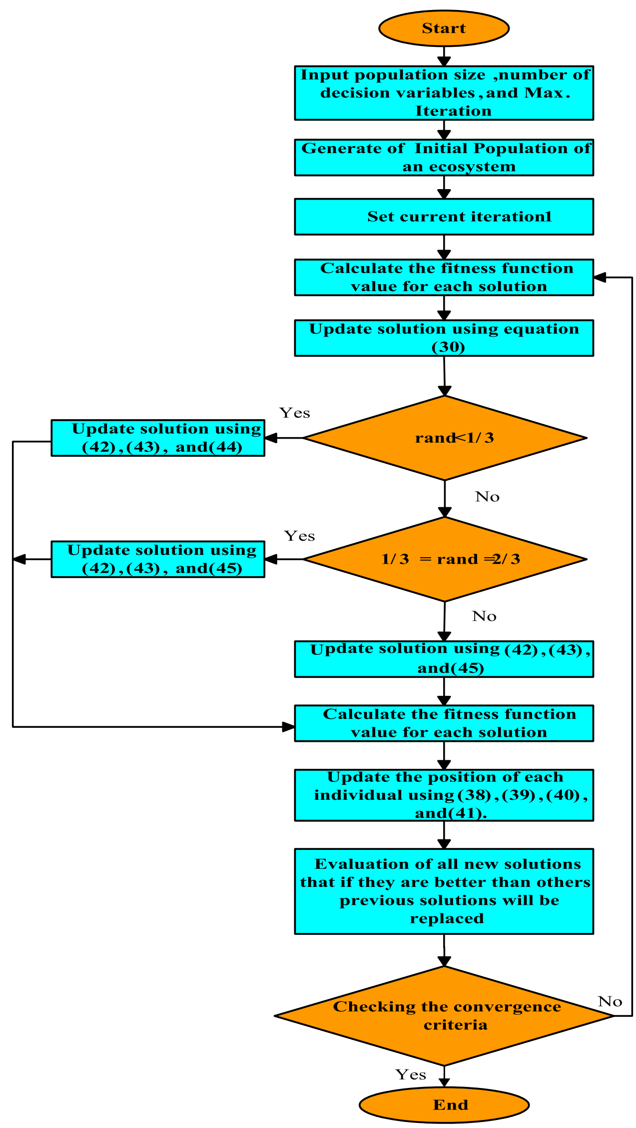

Fig. 4 Flowchart of the proposed IAEO algorithm.

two weight variables e and h. The decomposition improves the exploitation of the AEO algorithm [59].

The position of the i^{th} individual x_i in the population can be updated depending on the decomposer x_n and using a decomposition factor D and two weight variables e and h according to the following expression [59]:

$$x_i(t+1) = x_n(t) + D.(e.x_n(t) - h.x_i(t)), \quad i = 1, ..., n \tag{38}$$

$$D = 3u, \quad u\tilde{\mathrm{N}}(0,1) \tag{39}$$

$$e = r_3.randi([12]) - 1 \tag{40}$$

$$h = 2.r_3 - 1 \tag{41}$$

The flowchart describing the procedure of the conventional AEO algorithm is shown in Fig. 3.

5.2. IAEO

The IAEO improves the performance of the original AEO. The AEO performs in three phases (production, consumption, and decomposition), however, the IAEO is proposed to improve the performance of AEO in the consumption phase. The sine–cosine method has been applied instead of (35), (36), and (37). Where, sine- cosine function creates different solutions and fluctuates outwards or towards the best possible solution [39]. The improved equation can be expressed as follows:

$$\mathrm{r}_1 = 2 - It \times (\frac{2}{MaxIt}) \tag{42}$$

$$r_3 = (2 \times pi) \times rand\,(0,1) \tag{43}$$

where It is the current iteration, $MaxIt$ is the maximum number of iterations. and r_1 and r_3 are random numbers between (0,1) [39].

$$x_i(t+1) = \left\{ \begin{array}{c} x_i(t) + r_1 \times \sin(r_3) \times C \times (x_i(t) - x_1(t))r_4 \\ < 0.5 \quad i \in [2, ..., n] \\ x_i(t) + r_1 \times \cos(r_3) \times C \times (x_i(t) - x_1(t))r_4 \\ > 0.5 \; i \in [2, ..., n] \end{array} \right\} \tag{44}$$

$$\left\{ \begin{array}{c} x_i(t+1) = \left\{ \begin{array}{c} x_i(t) + r_1 \times \sin(r_3) \times C \times (x_i(t) - x_j(t))r_4 \\ < 0.5, \; i \in [3, ..., n] \\ x_i(t) + r_1 \times \cos(r_3) \times C \times (x_i(t) - x_j(t))r_4 \\ > 0.5, \; i \in [3, ..., n] \end{array} \right\} \\ j = randi([2i - 1]) \end{array} \right. \tag{45}$$

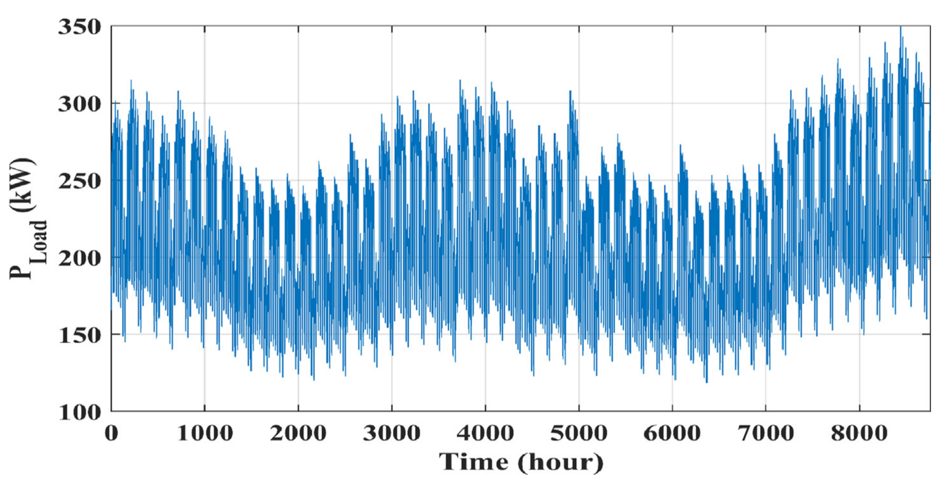

Fig. 5 Annual load curve at the proposed site of the hybrid system under study.

Table 2 Results of the statistical study for the three proposed hybrid systems using IAEO and AEO techniques.

	PV + Wind + FC		Wind + FC		PV + FC	
	IAEO	AEO	IAEO	AEO	IAEO	AEO
Best	1.90840	1.90896	1.40560	1.40591	1.35265	1.35378
Worst	1.93277	1.91596	1.43119	1.42008	1.36153	1.35922
Mean	1.91109	1.91154	1.41080	1.41197	1.35545	1.35597
Median	1.90984	1.91147	1.40953	1.41127	1.35516	1.35603
RE	0.38247	0.15857	0.45839	0.33939	0.21975	0.13362
MAE	0.07053	0.06775	0.18484	0.21583	0.10352	0.08114
SD	0.00269	0.00258	0.00519	0.00606	0.00280	0.00219
RMSE	0.00464	0.00302	0.00689	0.00693	0.00354	0.00256
Eff.	99.85952	99.86473	99.63268	99.57075	99.79362	99.83806

$$
\left\{ x_i(t+1) = \begin{cases} x_i(t) + r_1 \times \sin(r_3) \times C \\ \times (r_2(x_i(t) - x_j(t)) + (1 - r_2)(x_i(t) - x_j(t))), r_4 \\ < 0.5, i \in [3, \ldots, n] \\ x_i(t) + r_1 \times \cos(r_3) \times C \times (r_2(x_i(t) - x_j(t)) \\ +(1 - r_2)(x_i(t) - x_j(t))), r_4 > 0.5, i \in [3, \ldots, n] \end{cases} \right\} \\ j = randi([2i - 1]) \tag{46}
$$

The overall process of IAEO can be summarized in Fig. 4.

6. Case study

To validate the robustness and stability of the proposed IAEO technique, it has been applied for addressing the studied problem of the optimal configuration of the proposed PV/wind turbine/FC energy system. The proposed hybrid system has been introduced in the grid-connected and the islanded modes for satisfying the load requirements in the site. Ataka region, in Suez city in Egypt, on the Red Sea shore, located at 30.0° north latitude and 32.5° east longitude, is selected as the location of the case study. The site is rich in renewable energy resources described by the high solar radiation intensity and the continuous blow of wind at sufficient speed for operating the wind turbines. Egypt has experience in installing wind farms in the selected region along the Suez Gulf, a 380 MW wind farm in Gabal Elzeit entered the commercial operation since 2015. From the PV power potential over the territory of Egypt, it is obvious that the area along the Suez Gulf is very rich with solar energy [14]. From the atlas of wind resources in Egypt, it is noticed that the Suez Gulf region has good wind power generation conditions where the average wind speed in this region is about 10.5 m/s [1]. The annual load curve of Ataka region over one hour is shown in Fig. 5. Two types on the hybrid system, grid-connected and standalone systems, are evaluated for meeting the load demand in the site. In each type, three different configurations of the hybrid system are examined. These configurations are: i) PV/wind/FC system, ii) wind/FC system, iii) PV/FC system. The proposed IAEO is validated in determining the optimal sizes of these six hybrid systems and the optimization results are comprehensively compared with the corresponding ones obtained by the conventional AEO.

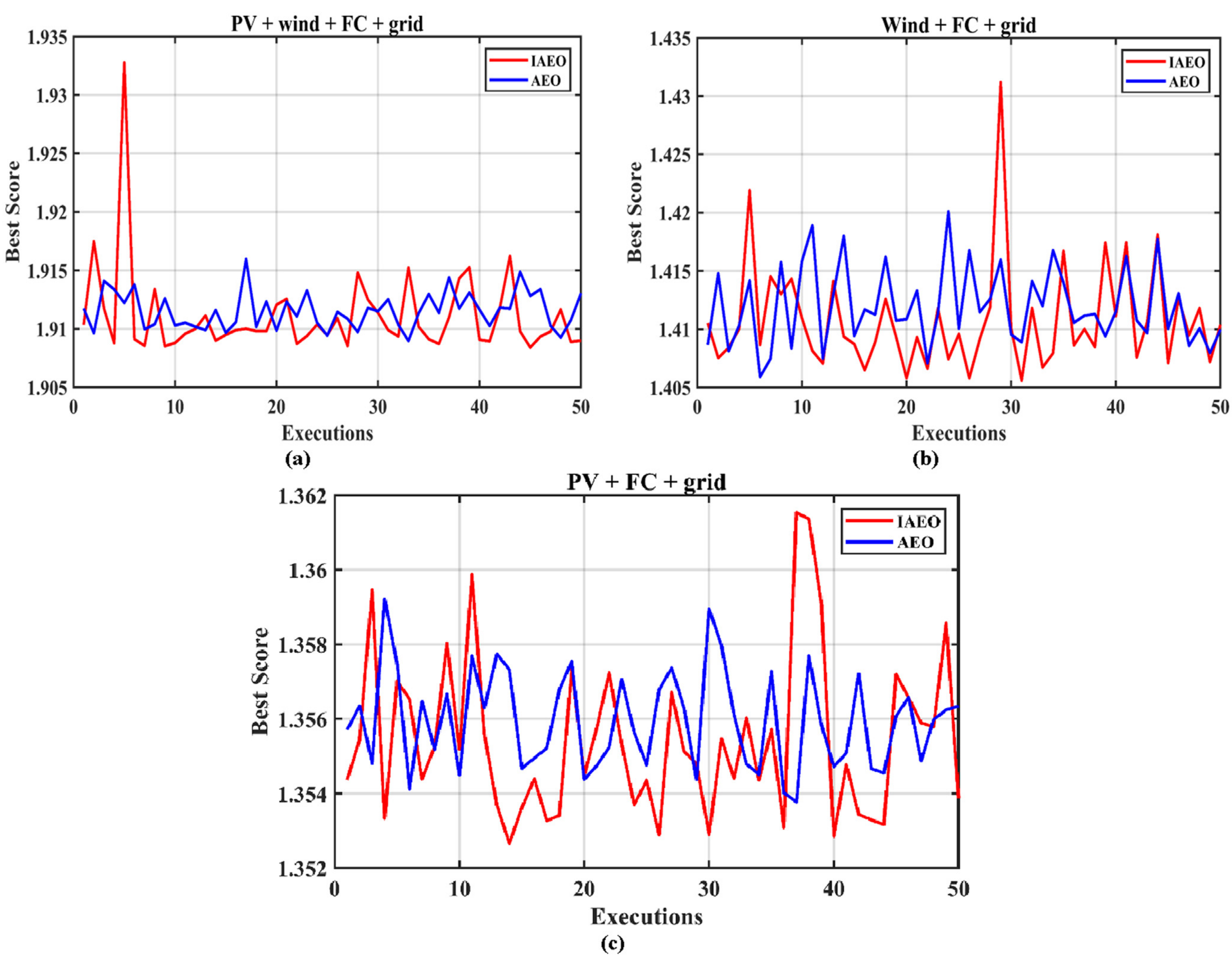

Fig. 6 End values of the objective function of the 50 individual run using IAEO and AEO: (a) PV/WT/FC system, (b) WT/FC system, (c) PV/FC system.

7. Simulation results and discussion

The main object of this paper is to generate the optimal design of the proposed hybrid system and validate the accuracy of the proposed IAEO technique. The optimal sizing is based on the objective functions introduced in (23) and (25) and the parameters of optimization are: **i)** the number of PV modules N_{PV}, **ii)** the wind turbines number included in the wind farm N_{WT}, **iii)** the rated value of water electrolyzer power P_{Elect_rated}, **iv)** the mass of the hydrogen gas tank $M_{tank,\ \max}$, **v)** the rated value of the FC power P_{FC_rated}, vi) the rated power of the inverter P_{Inv_rated}. IAEO and AEO are launched 50 times for each configuration and a statistical study has been conducted based on a set of measures; the best minimum value of the fitness function, the worst maximum value, the mean value of the 50 obtained values of the objective function and the median. For a deep analysis of the obtained results and to ensure the sensitivity and stability of the proposed algorithm, relative error (RE), standard deviation (SD), and root mean square error (RMSE) have been calculated based on the results of the 50 executions. In the next subsections, the results of the grid-connected and standalone systems are provided. Modelling and simulation of the optimization problem have been accomplished using MATLAB 2018a program. The results obtained by the AEO and IAEO are compared with those obtained by PSO [61], SSA [61,62], and GWO [63,64].

For AEO and IAEO algorithms, the number of runs (N_{run}) = 50, maximum iterations = 100 iterations, and the population size = 20.

For PSO, N_{run} = 50, population size = 20, maximum iteration = 100, c_1 = 2, c_2 = 2, w_{max} = 0.9, w_{min} = 0.2.

For SSA, N_{run} = 50, search agents = 20, maximum iteration = 100.

For GWO, N_{run} = 50, search agents = 20, maximum iteration = 100, a linearly decreased from 2 to 0.

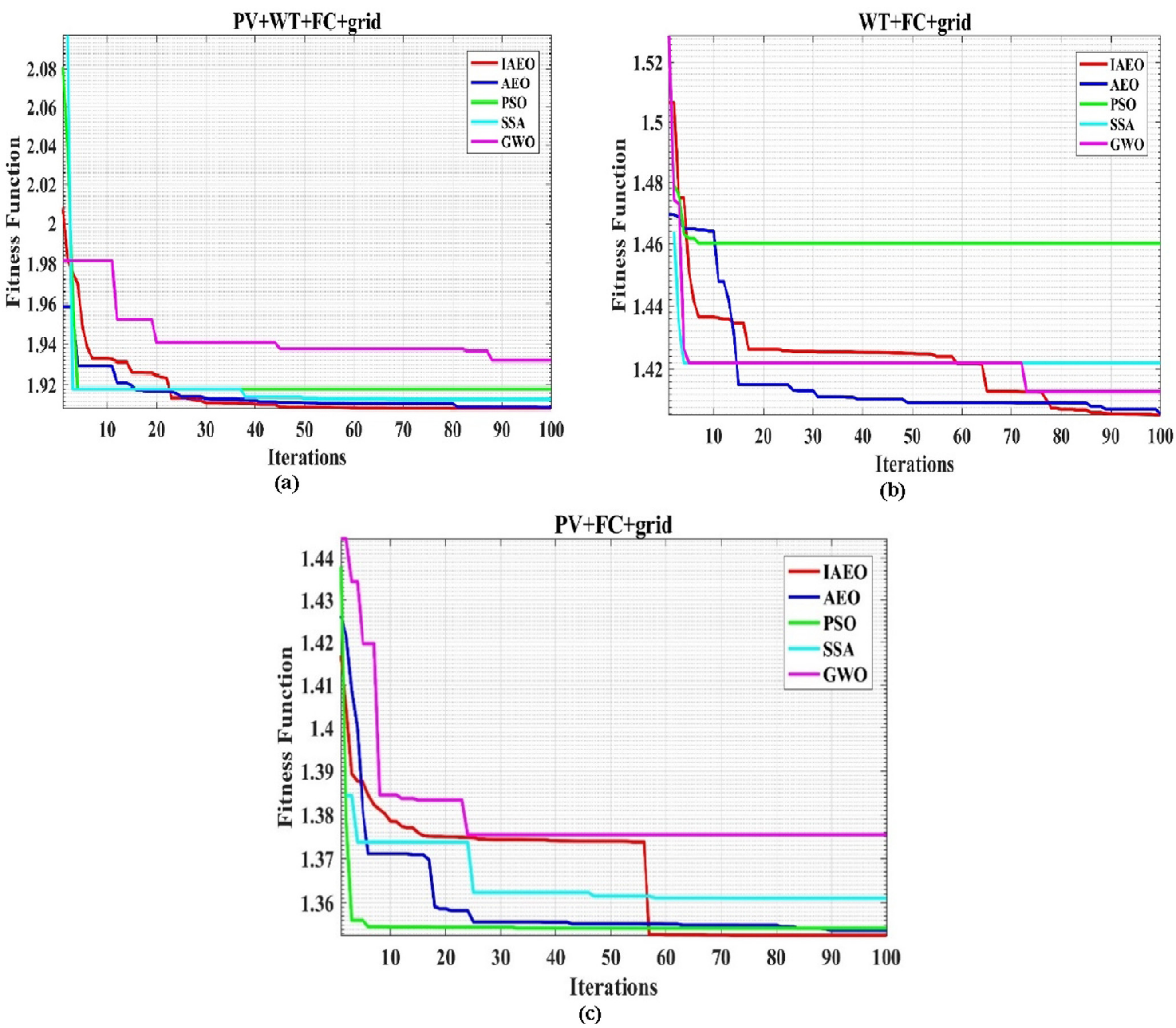

Fig. 7 Convergence curves for different optimization algorithms: (a) PV/WT/FC system, (b) PV/WT/FC system, (c) PV/WT/FC system.

Table 3 Optimization results of the system components based on IAEO and other algorithms for PV/WT/FC hybrid system integrated into the grid.

		IAEO	AEO	PSO	SSA	GWO
	Best fitness function	1.90840	1.90896	1.918023	1.912848	1.931887
Best solution	PV (units)	260.0132	260.0182	260	260.000	260.0000
	Wind (units)	60.0152	60.0111	60	60.0000	60.0000
	Electrolyzer (kW)	710.0505	712.0299	710	724.7619	710.0000
	Hydrogen tank (kg)	129.0353	130.8610	200	157.4691	158.1626
	Fuel cell (kW)	200.0237	200.1415	200	200.0000	23.8356
	DC/AC converter (kW)	400.2782	401.0518	400	400.9362	424.9118
	Computation time (s)	110.853	165.913	45.835	631.817	116.177
	COE ($/kWh)	0.41243	0.41304	0.422926	0.417422	0.437675
	NPC ($)	9,957,537	9,972,230	10,210,924	10,078,019	10,567,013
	LPSP	2.9194×10^{-18}	1.8961×10^{-18}	9.781×10^{-19}	1.203×10^{-18}	4.364×10^{-19}
	D_load	1.3153	1.3153	1.31506	1.31506	1.31506
	D_gs	35.7710	35.7706	35.7659	35.7659	35.7659
	STD	56.8350	56.8344	56.8255	56.8255	56.8255

Table 4 Optimization results of the system components based on IAEO and other algorithms for WT/FC hybrid system integrated into the grid.

		IAEO	AEO	PSO	SSA	GWO
	Best fitness function	1.40560	1.40591	1.460145	1.421900	1.4129431
Best solution	PV (units)	–	–	–	–	–
	Wind (units)	90.0088	90.0088	90.0000	90	90
	Electrolyzer (kW)	410.0842	410.0842	410.0000	410	428.3915
	Hydrogen tank (kg)	130.6993	130.6993	124.4495	100	129.8923
	Fuel cell (kW)	320.4715	320.4715	400.0000	320	327.8084
	DC/AC converter (kW)	402.3633	402.3633	400.0000	400	400
	Computation time (s)	142.139	126.011	42.459	704,713	85.900
	COE ($/kWh)	0.45004	0.45023	0.529527	0.467430	0.457901
	NPC ($)	10,865,694	10,870,047	12,784,625	11,285,398	11,055,336
	LPSP	3.00971×10^{-20}	3.00971×10^{-20}	0	0	3×10^{-20}
	D_load	0.86535	0.86539	0.865336	0.865336	0.865336
	D_gs	16.0056	16.0072	16.0051	16.0051	16.0051
	STD	40.2598	40.2657	40.2579	40.2579	40.2579

Table 5 Optimization results of the system components based on IAEO and other algorithms for PV/FC hybrid system integrated into the grid.

		IAEO	AEO	PSO	SSA	GWO
	Best fitness function	1.35265	1.35378	1.3542653	1.361034	1.375417
Best solution	PV (units)	240.000	240.01787	240.0000	240.000	240.0000
	Wind (units)	–	–	–	–	–
	Electrolyzer (kW)	500.0070	504.5705	500.0000	500.000	500.0000
	Hydrogen tank (kg)	128.9673	130.8601	128.8653	147.730	113.8111
	Fuel cell (kW)	300.0055	300.2831	300.0000	300.000	300.0000
	DC/AC converter (kW)	400.1254	400.0174	400.0000	494.953	402.2744
	Computation time (s)	107.487	146.201	42.504	514.645	100.654
	COE ($/kWh)	0.39029	0.39135	0.416043	0.39921	0.414512
	NPC ($)	9,423,136	9,448,473	10,044,733	9,638,349	10,007,762
	LPSP	0	2.40777×10^{-19}	0	2.407×10^{-19}	0
	D_load	0.92802	0.92809	0.928022	0.92802	0.928022
	D_gs	19.3635	19.3655	19.3635	19.3635	19.36351
	STD	38.6790	38.6847	38.6790	38.6790	38.67901

7.1. Grid-connected hybrid system

7.1.1. Validation of iaeo algorithm

The results of the statistical study performed based on the final values of the fitness function of the 50 individual executions of the optimization problem for each configuration using the proposed IAEO and the conventional AEO algorithms are listed in Table 2. From this table, it can be observed that the IAEO successes to reach the optimal fitness function value in all cases. The insignificant values of SD, RE, and RMSE proved the convergence between the obtained values in the 50 implements, which proved the stability and accurateness of the IAEO technique. The end values of the objective function over the 50 implements presented in the graphical form for the three studied configurations using the two algorithms, i.e., IAEO and AEO, are shown in Fig. 6. The reader can obviously notice from Fig. 6 that the values of the fitness function for the proposed IAEO occurred in a narrow range, which proved the stability of the proposed algorithm. Consequently, the values of the parametric and nonparametric metrics for the proposed IAEO are better than those obtained by the original AEO.

7.1.2. Optimal design of the studied system

In this subsection, the implements of the optimization program that results in the optimal value of the fitness function in all cases are highlighted. The convergence curves of the proposed IAEO algorithm are compared with those obtained from the conventional AEO algorithm as well as other well-known algorithms, PSO, SSA, and GWO. The convergence curves corre-

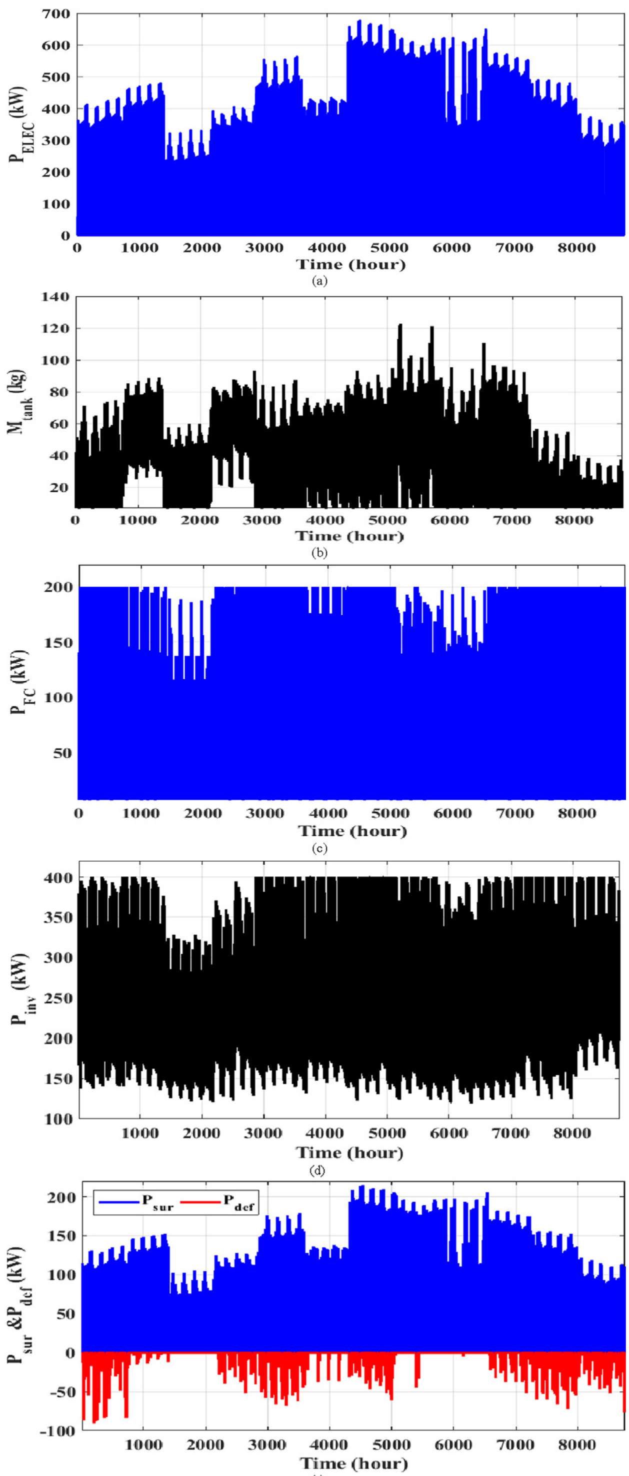

Fig. 8 Annual operation of PV/WT/FC grid-connected system based on the results obtained by IAEO: (a) Electrolyzer output power, (b) Mass of Hydrogen stored, (c) fuel cell output power, (d) inverter power, (e) surplus and deficit power.

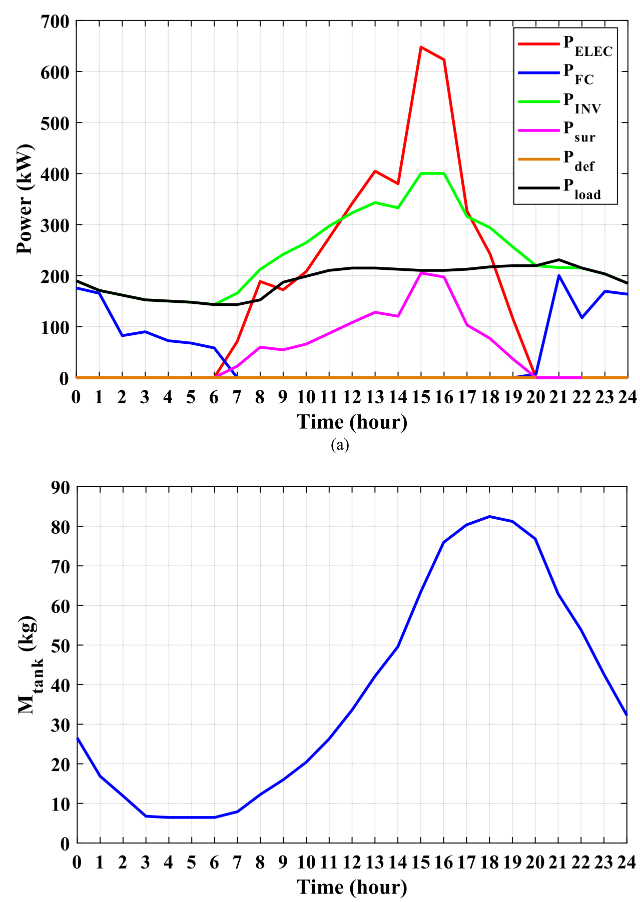

Fig. 9 Operation of the PV/WT/FC grid-connected system on a certain day in the summer (24 h) using the results of IAEO: (a) Operating power for each subsystem, (b) Mass of hydrogen stored in the tank.

sponding to these algorithms are shown in Fig. 7. From this figure, the reader can notice that the fitness function value obtained from the IAEO algorithm is better than the other algorithms in all proposed configurations. The results of the optimization parameters, i.e. (N_{PV}, N_{WT}, P_{Elect_rated}, $M_{tank,\max}$, P_{FC_rated}, P_{Inv_rated}) for all studied configurations are listed in Table 3, 4 and 5. Based on these tables, the reader can notice that using both IAEO and AEO algorithms, the best minimum fitness function values are for the third configuration, i.e. hybrid PV/FC power system integrated with the external grid. For this configuration, the proposed IAEO generates the objective function of **1.35265** where AEO generates an objective function equals **1.3537782**. The value of COE obtained from that system equals **0.3902975** \$/kWh for IAEO and **0.3913469** \$/kWh for AEO. The smaller value of COE in the case of using IAEO results in the least value of the NPC of \$**9423136.57**. For PV/WT/FC hybrid system, the annual cost of energy purchased from the external grid is **\$907.79,** while

Table 6 Techno-economic characteristics of system components for the optimal case for PV/WT/FC hybrid system integrated to the grid.

	PV	Wind	Electrolyzer	FC	Inverter
Capital cost (\$)	1820092.194	1164295.103	1420100.975	600071.207180739	320222.554
O&M cost (\$/year)	5200.263	4501.141	17751.262196	35004.1537522098	3202.225
Replacement cost (\$)	0	900228.2	1,065,076	500059.3	300208.6
Rated Capacity (kW)	260.0132	450.114	710.05	200.0237	400.278
Mean output (kW)	231.0173	129.546	132.191	41.270	255.445
Total production (kWh/year)	2023711.300	1134829.934	1157997.613	361528.316	2237706.406

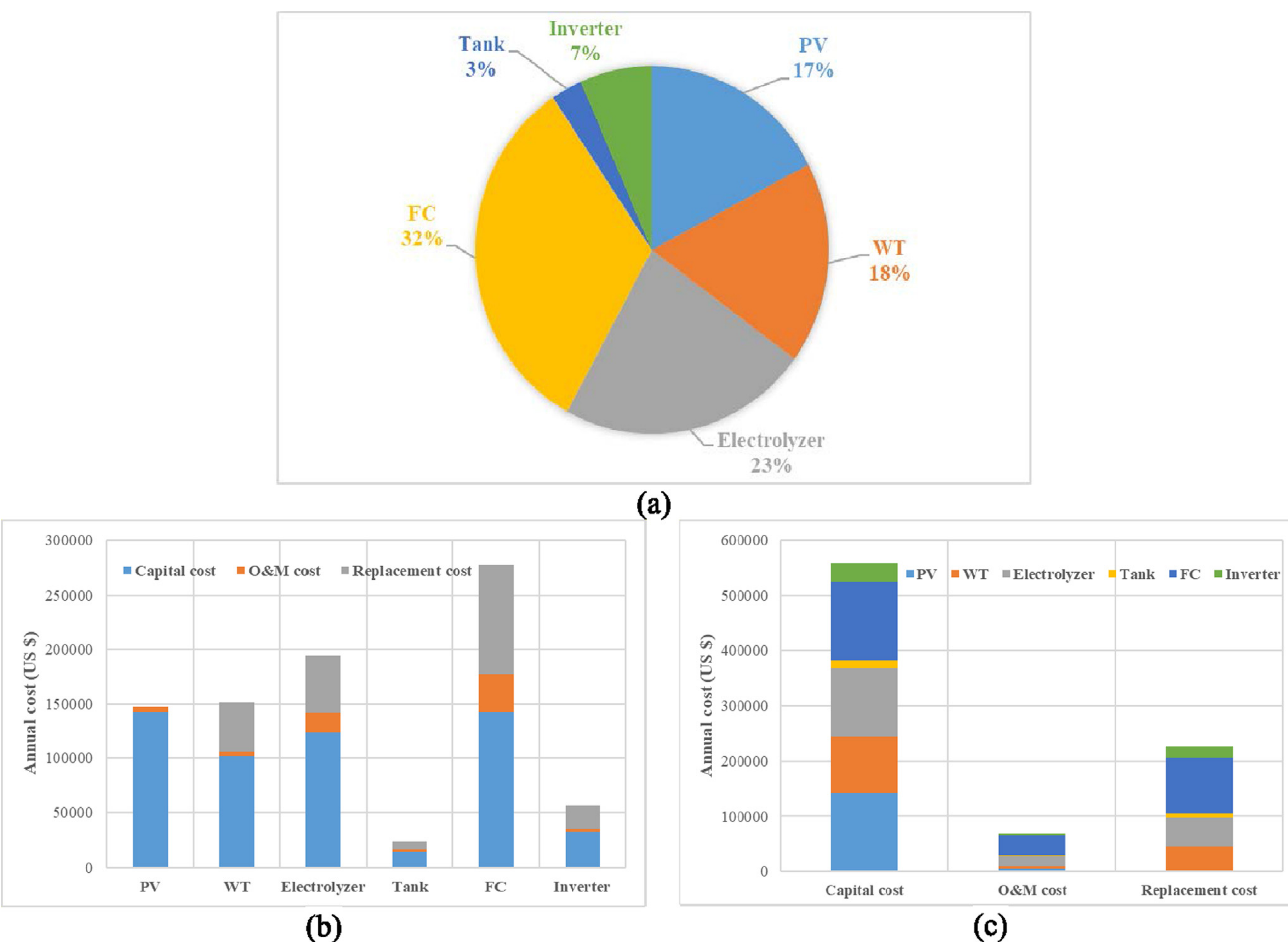

Fig. 10 Contribution of the system component in the total cost: (a) NPC contribution for each subsystem, (b) Breakdown of the annual system cost regarding technology, (c) Breakdown of the annual system cost regarding the type of cost.

Table 7 Results of the statistical study for the three proposed hybrid configurations using IAEO and AEO techniques.

	PV + Wind + FC		Wind + FC		PV + FC	
	IAEO	**AEO**	**IAEO**	**AEO**	**IAEO**	**AEO**
Best	0.23868	0.23872	0.24853	0.24887	0.31713	0.31714
Worst	0.24011	0.23919	0.25310	0.25227	0.31794	0.31765
Mean	0.23892	0.23886	0.25060	0.24958	0.31732	0.31731
Median	0.23882	0.23883	0.25024	0.24949	0.31726	0.31730
RE	0.02939	0.01151	0.13148	0.05370	0.01602	0.01278
MAE	0.05176	0.03143	0.41606	0.14458	0.02909	0.02817
SD	0.00024	0.00015	0.00206	0.00071	0.00018	0.00017
RMSE	0.00038	0.00018	0.00244	0.00089	0.00024	0.00021
Eff.	99.89673	99.93719	99.17740	99.71211	99.94186	99.94370

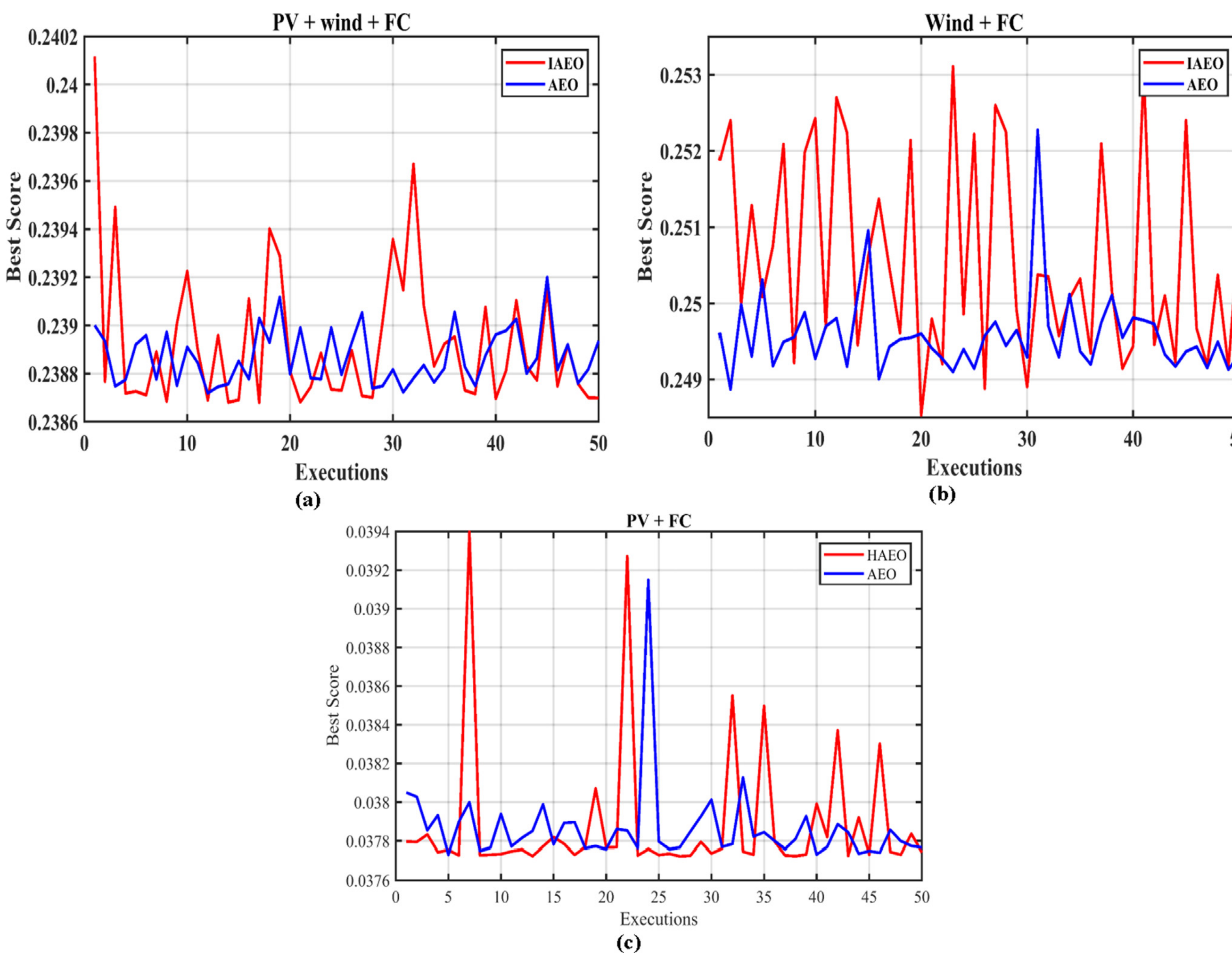

Fig. 11 End values of the objective function for 50 executions using IAEO and AEO: (a) PV/WT/FC system, (b) WT/FC system, (c) PV/FC system.

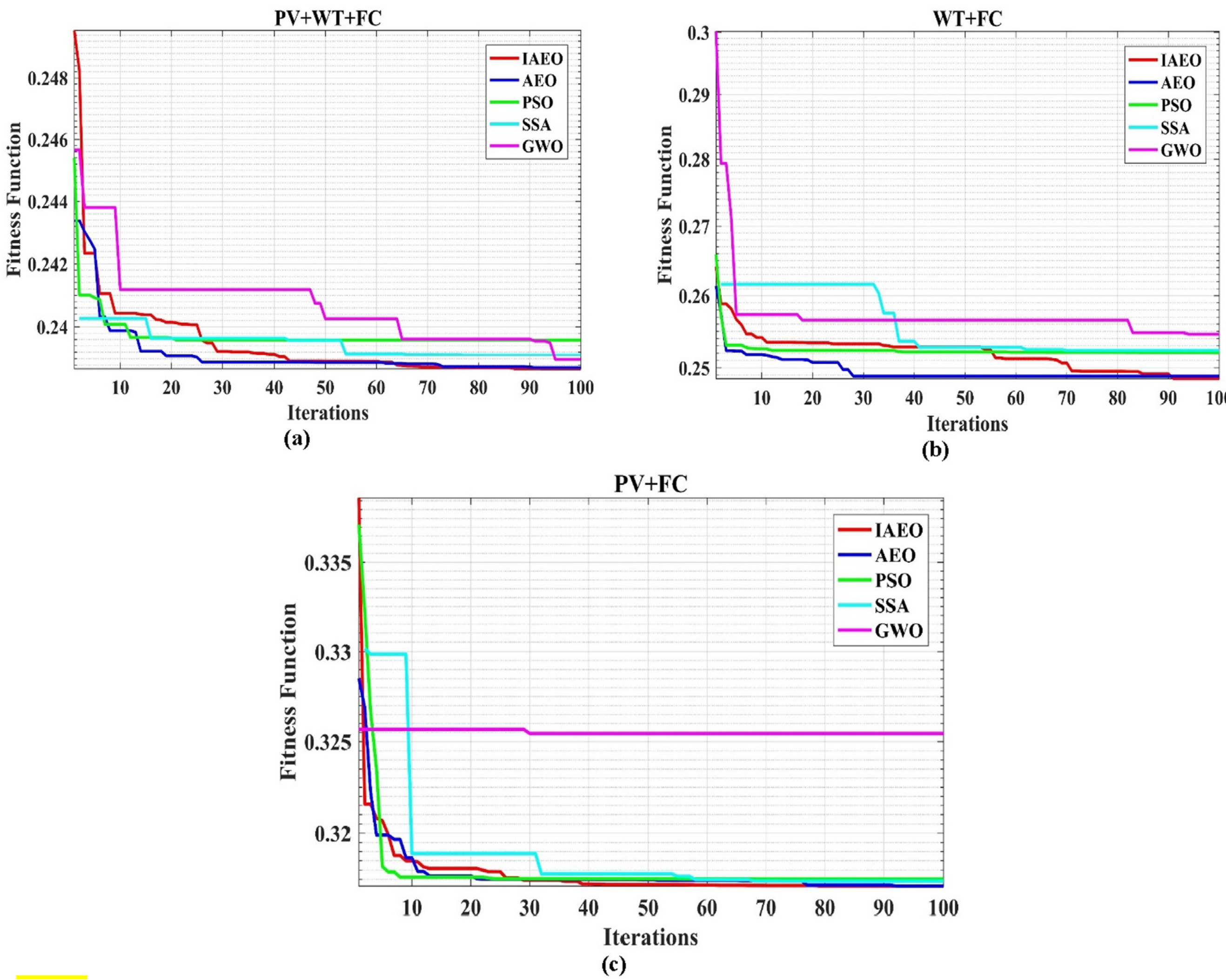

Fig. 12 Convergence curves for different optimization algorithms: (a) PV/wind/FC system, (b) PV/wind/FC system, (c) PV/wind/FC system.

Table 8 Results of the system components based on IAEO and other algorithms for PV/WT/FC hybrid isolated system.

		IAEO	AEO	PSO	SSA	GWO
Best fitness function		0.23868	0.23872	0.23958	0.23910	0.23897
Best solution	**PV (units)**	263.2427	266.4372	300.000	260.535	262.584
	Wind (units)	69.6946	66.8851	60.0000	71.5228	68.6599
	Electrolyzer (kW)	710.0031	710.2769	710.000	710.000	710.000
	Hydrogen tank (kg)	103.1773	103.3155	103.391	108.266	107.143
	Fuel cell (kW)	200.0021	200.0114	200.000	200.000	200.000
Computation time (s)		78.6674	87.698	33.918	309.678	76.496
COE ($/kWh)		0.42957	0.43229	0.43344	0.43442	0.43411
NPC ($)		10,437,193	10,371,455	10,392,504	10,488,564	10,408,633
LPSP		0.02449	0.02794	0.03385	0.02309	0.02692
E_dummy (kWh)		434,630	428,917	477,739	434,317	426,899

the annual cost of energy sold to the grid is **$73339.84.** The computation time required for each algorithm for solving the optimization problem is also provided in Tables 3, 4, 5. Except for the PSO algorithm, the time taken for performing the proposed IAEO-based optimization program is the least within all optimization methods included in the study for all grid-connected configurations. For WT/FC hybrid system, the annual cost of energy purchased from the external grid is **$11414.6** while the annual cost of energy sold to the grid is **$25769.75**. For PV/FC hybrid system, the annual cost of energy purchased from the external grid is **$1083.69** while the annual cost of energy sold to the grid is **$33969.71**. The reason behind the smallest value of the objective function and in turn the COE in the PV/FC hybrid system can be clarified by the very high initial and replacement costs of the wind turbines, which included in the other two configurations. From Table 3, 4, and 5, it is obvious that IAEO generates the minimum value of COE in all cases and at the same time ensures the low value of the LPSP below the predefined value, which validates the high reliability of the proposed hybrid systems.

Table 9 Results of the system components based on IAEO and other algorithms for WT/FC hybrid isolated system.

		IAEO	AEO	PSO	SSA	GWO
Best fitness function		0.24853	0.24887	0.25206	0.252355	0.2545941
	PV (units)	–	–	–	–	–
	Wind (units)	217.9351	216.30845	230.377	232.9933	243.9508
Best solution	**Electrolyzer (kW)**	887.3425	881.3575	880.000	880.7045	889.5225
	Hydrogen tank (kg)	158.8516	151.5210	200.000	184.3122	184.8468
	Fuel cell (kW)	300.0290	300.1877	300.000	300.0495	300.0000
Computation time (s)		72.1345	93.769	37.540	383.902	85.569
COE ($/kWh)		0.65182	0.65548	0.67507	0.677132	0.6930300
NPC ($)		15,825,564	15,737,413	16,298,619	16,348,334	16,732,150
LPSP		0.04922	0.05112	0.04579	0.042241	0.0347072
E_dummy (kWh)		751,537	758,896	768,749	821,912	911,294

Table 10 Results of the system components based on IAEO and other algorithms for PV/FC hybrid isolated system.

		IAEO	AEO	PSO	SSA	GWO
Best fitness function		0.31713	0.31714	0.3175090	0.3173993	0.325450
	PV (units)	600.03821	600.00480	600.0000	600.0000	611.6447
	Wind (units)	–	–	–	–	–
Best solution	**Electrolyzer (kW)**	705.6529	701.4865	700.0000	700.6573	762.6199
	Hydrogen tank (kg)	166.4268	165.8928	165.6353	168.8295	228.4437
	Fuel cell (kW)	303.6263	300.0689	300.0000	300.0000	302.0816
Computation time (s)		61.449	82.0939	32.056	406.981	77.431
COE ($/kWh)		0.51915	0.519780	0.522186	0.5192796	0.539254
NPC ($)		12,549,496	12,534,308	12,557,217	12,537,213	13,019,469
LPSP		0.056334	0.056370	0.0465422	0.0540839	0.051802
E_dummy (kWh)		1,085,419	1,091,395	1,093,789	1,092,469	1,058,099

Table 11 Techno-economic characteristics of system component for the optimal case for PV/WT/FC hybrid system integrated to grid.

	PV	Wind	Electrolyzer	FC	Inverter
Capital cost ($)	1842698.791	1352075.634	1420006.145	600006.402	280,000
O&M cost ($/year)	5264.853	5227.096	17750.076	35000.373	2800
Replacement cost ($)	0	1,045,419	1,065,005	500005.3	262,500
Rated Capacity (kW)	263.242	522.67	710.003	200.002	350
Mean output (kW)	233.886	150.440	145.686	32.360	215.60
Total production (kWh/year)	2048846.965	1317858.248	1276215.878	283480.319	1888664.315

7.1.3. Operation of the studied system under optimized parameters

In this subsection, the results of Table 3 for the PV/wind/FC system are applied to the proposed system model and the behavior of the studied system has been emphasized. Fig. 8 shows the electrical power output from the electrolyzer, the stored mass of hydrogen gas, the output power produced from the FC, and the AC power supplied by the inverter. For more understanding of Fig. 8, the results are focused on a certain summer day, i.e. 24 h of studied system operation. Fig. 9 shows the same results provided in the previous figure but on 24 h' basis. At night and early hours of the day, when the electrical power produced from the PV system is missed or not sufficient, the shortage in production is covered by supplying the FC with the stored hydrogen in the tank, which results in a decrease in the mass of hydrogen in the tank. As the power generated from the PV system is gradually increased, the electrical power provided by the FC is decreased until the power supplied by the renewable sources is sufficient to feed the load requirements. Starting from that point, the electrolyzer will convert the excess electrical energy into the chemical form, i.e., the stored mass of hydrogen gas will be increased as shown in Fig. 9(b). When the power produced from the PV and WT power plants exceeds the load demand plus the rated electrical power of the water electrolyzer, the excess power is supplied to the external national grid.

Table 6 summarizes the techno-economic characteristics of the stand-alone PV/WT/FC hybrid system for the optimal case. The contribution of each subsystem to the NPC of the proposed hybrid system in the optimal case is provided in Fig. 10(a). The NPC of the PV subsystem accounts for 17% of the NPC of the whole system. The NPC of the WT is representing 18% of NPC os the system. The NPC of the electrolyzer, hydrogen tank, FC, and inverter is representing 23%, 32%, 3%, and 7% of the system NPC, respectively. Fig. 10(b) and 10(c) show the contribution of the all costs of each component to the annual cost of the hybrid system and inturns the NPC of the system. In Fig. 10(b) the annual costs are broken regarding the generating technology, while in Fig. 10(b) the annual costs are presented regarding the type of cost. From Fig. 10(b) it can be noticed that the FC system has the highest annual cost within all subsystems. From Fig. 10 (c) it can be noticed that the FC system has the highest values of all annual costs; annual interest of capital cost ($142454.74), operating and maintenance costs ($35,004.15) and annual replacement cost of ($100,011.9).

7.2. Standalone hybrid system

7.2.1. Validation of IAEO algorithm

The results of the statistical measurements for the proposed IAEO and the conventional AEO algorithms are listed in

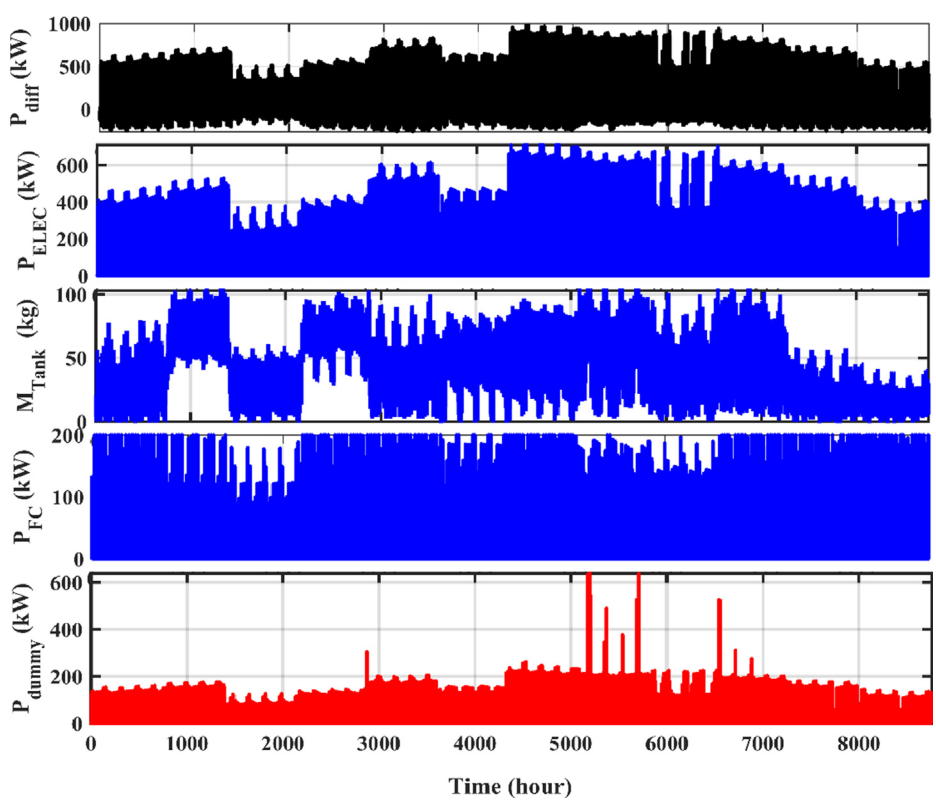

Fig. 13 Annual operation of the PV/WT/FC isolated hybrid system using the results of IAEO.

Table 7. From Table 7, the reader can conclude that the IAEO generates the best minimum value of the fitness function in all cases. The insignificant values of the statistical metrics involved in the study proved the sensitivity and stability of IAEO. The final values of the objective function of 50 individual launches of the optimization program using the two algorithms, i.e. IAEO and AEO, are shown in Fig. 11.

7.2.2. Combinations of the studied system components

This subsection presents the results obtained in the execution that results in the best fitness function value using both IAEO and AEO techniques. The convergence curves of AEO, IAEO, PSO, SSA, and GWO algorithms for the three possible combinations are shown in Fig. 12. The optimized parameters (i.e. N_{PV}, N_{WT}, P_{Elect_rated}, $M_{tank,\max}$, P_{FC_rated}) for the three suggested combinations are listed in Table 8, 9, and 10. In the case of the isolated system, the rating of the inverter is taken as the value of the peak load. From this table, the reader can notice that using both IAEO and AEO algorithms, the best minimum values of fitness function are for the first configuration, i.e. hybrid PV/wind/FC power system. It is obvious that IAEO generates the minimum value of COE in all cases and at the same time ensures the low value of the LPSP below the predefined value, which validates the high reliability of the proposed hybrid systems.Table 11.

7.2.3. Operation of hybrid system under the optimized parameters

In this section, the results obtained in Table 7 for the PV/WT/FC standalone system are applied in the mathematical model of the system and the operation of the studied system is highlighted. Fig. 13 shows the annual operation of the hybrid standalone system. For more understanding of Fig. 13, the results are presented only for 24 h, i.e. one-day operation. Fig. 14 shows the following characteristics; the difference between the renewable generation and demand of load, the power output from the water electrolyzer, the stored mass of hydrogen gas in each hour, the power output from the FC system, and the power consumed in the dummy load in the case of excess power from PV and WT sources.

Table 9 summarizes the techno-economic characteristics of the stand-alone PV/WT/FC hybrid system for the optimal case. The contribution of each subsystem to the NPC of the proposed hybrid system in the optimal case is provided in Fig. 15(a). The NPC of the PV subsystem accounts for 17% of the NPC of the whole system. The NPC of the WT is representing 20% of NPC os the system. The NPC of the electrolyzer, hydrogen tank, FC, and inverter is representing 23%, 32%, 2%, and 6% of the system NPC, respectively. Fig. 15(b) and 15(c) show the contribution of the all costs of each component to the annual cost of the hybrid system and inturns the NPC of the system. In Fig. 15(b) the annual costs are broken regarding the generating technology, while in Fig. 15(b) the annual costs are presented regarding the type of cost. From Fig. 15(b) it can be noticed that the FC system has the highest annual cost within all subsystems. From Fig. 15 (c) it can be noticed that the annual interest of the capital cost of the PV system has the highest value ($144,148.27), the FC

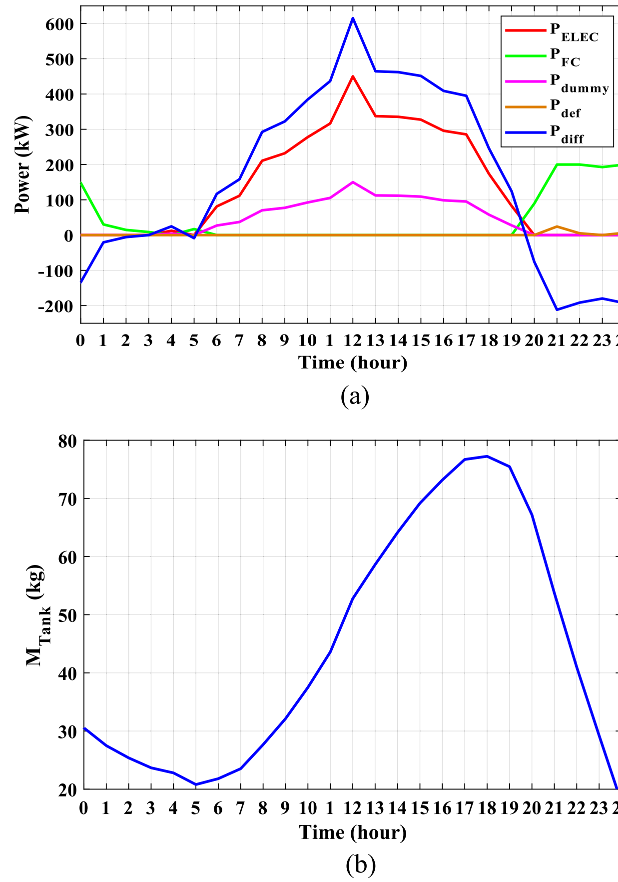

Fig. 14 Operation of the PV/WT/FC isolated hybrid system on a certain day in the summer (24 h) using the results of IAEO: (a) Operating power for each subsystem, (b) Mass of hydrogen stored in the tank.

system has the highest operating and maintenance costs ($35,000.37) and the highest annual replacement cost of ($100,001.1).

8. Conclusion

In this paper, the optimal configuration and component sizing of a proposed hybrid system has been implemented through the 25-year lifespan. The proposed hybrid system consists of PV, WT, FC, water electrolyzer, a tank of hydrogen gas as an energy storage system, and an inverter that facilitates the delivery of electrically generated energy to consumers. This hybrid system has been studied in two different cases (grid-connected and off-grid). This proposed system has been addressed as a complex optimization problem. A novel IAEO algorithm has been proposed and utilized to solve this problem and find the best optimal values of the subsystems included in the whole system under different configurations. The proposed IAEO has been tested on six different configurations in order to evaluate and validate its effectiveness and suitability in solving the studied optimization problem. The

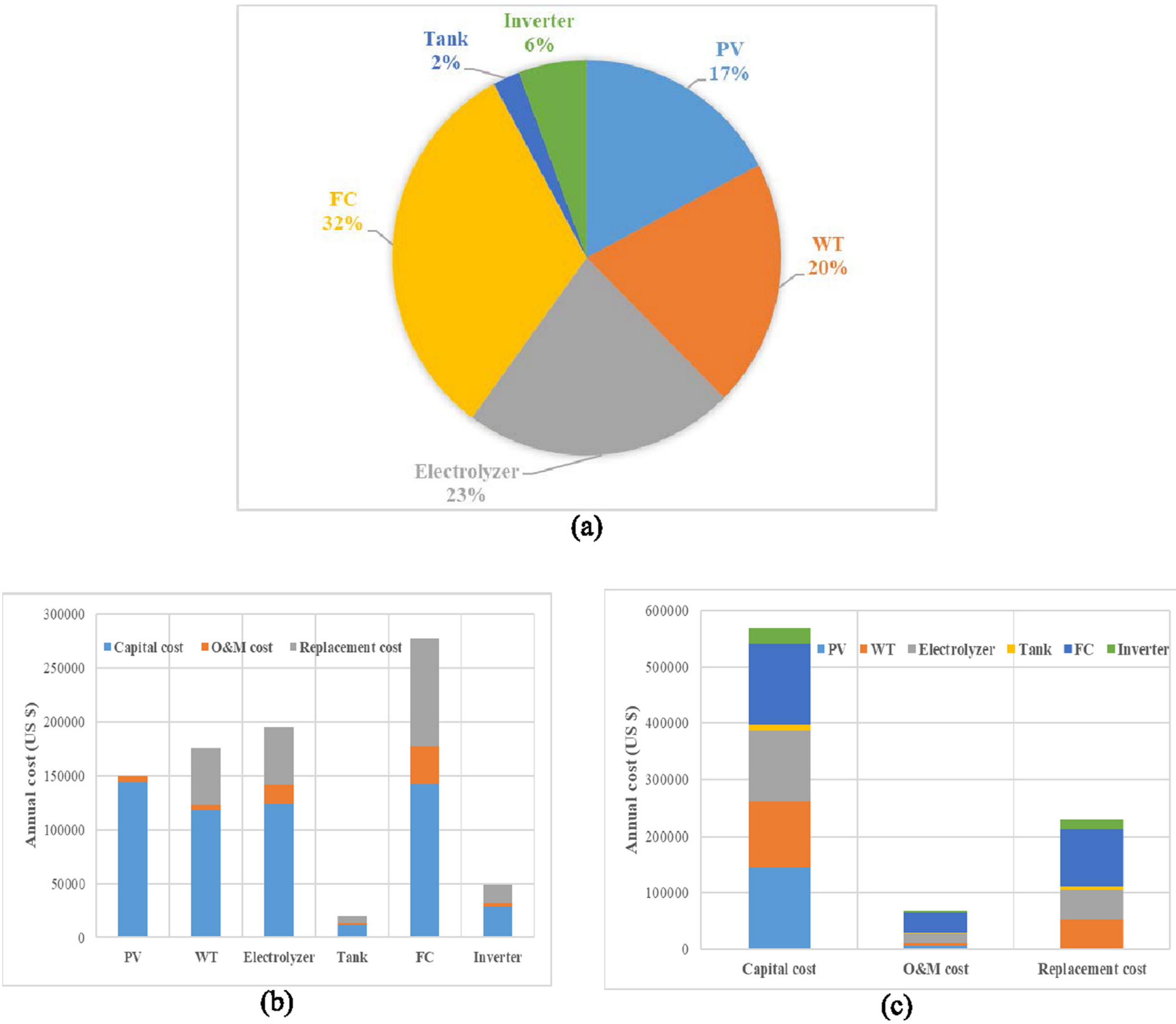

Fig. 15 Contribution of the system component in the total cost: (a) NPC contribution for each subsystem, (b) Breakdown of the annual system cost regarding technology, (c) Breakdown of the annual system cost regarding the type of cost.

IAEO algorithm provides the best results compared with those obtained by the conventional AEO, PSO, SSA, and GWO algorithms. The proposed IAEO algorithm provides fast convergence characteristics, the best minimum values of the objective function, and minimum COE. Based on the optimal configuration of the hybrid systems, it is found that the FC system has the highest contribution to the NPC. For the PV/WT/FC grid-connected configuration, it was found that the PV system has the highest annual interest of the capital cost and FC has the highest operating and maintenance and replacement cost, while for the PV/WT/FC off-grid configuration the FC has the highest cost all over. The proposed IAEO algorithm proved its suitability for solving the considered optimization problem over other optimization algorithms. In the future, the proposed IAEO could be used to solve other complex and non-linear engineering problems. In addition, the proposed system could be applied for different areas.

Declaration of Competing Interest

The authors declared that there is no conflict of interest.

References

[1] A.A.Z. Diab, H.M. Sultan, O.N. Kuznetsov, Optimal sizing of hybrid solar/wind/hydroelectric pumped storage energy system in Egypt based on different meta-heuristic techniques, Environ. Sci. Pollut. Res. (2019) 1–23.
[2] A.A.Z. Diab, H.M. Sultan, I.S. Mohamed, O.N. Kuznetsov, T. D. Do, Application of different optimization algorithms for optimal sizing of PV/wind/diesel/battery storage stand-alone hybrid microgrid, IEEE Access 7 (2019) 119223–119245.
[3] A. Evans, V. Strezov, T.J. Evans, Assessment of sustainability indicators for renewable energy technologies, Renew. Sustain. Energy Rev. 13 (5) (2009) 1082–1088.
[4] B. Jyoti Saharia, H. Brahma, N. Sarmah, A review of algorithms for control and optimization for energy management of hybrid

renewable energy systems, J. Renewable Sustainable Energy 10 (5) (2018) 053502.
[5] K. Kusakana, H.J. Vermaak, Hybrid renewable power systems for mobile telephony base stations in developing countries, Renewable Energy 51 (2013) 419–425.
[6] T. Ma, H. Yang, L. Lu, A feasibility study of a stand-alone hybrid solar–wind–battery system for a remote island, Appl. Energy 121 (2014) 149–158.
[7] H.M. Sultan, O.N. Kuznetsov, A.S. Menesy, S. Kamel, Optimal Configuration of a Grid-Connected Hybrid PV/Wind/Hydro-Pumped Storage Power System Based on a Novel Optimization Algorithm, in: 2020 International Youth Conference on Radio Electronics, Electrical and Power Engineering (REEPE), IEEE, 2020, pp. 1–7.
[8] P. Asmus, Why Microgrids Are Moving into the Mainstream: Improving the efficiency of the larger power grid, IEEE Electrif. Mag. 2 (1) (2014) 12–19.
[9] J.L. Bernal-Agustín, R. Dufo-Lopez, Simulation and optimization of stand-alone hybrid renewable energy systems, Renew. Sustain. Energy Rev. 13 (8) (2009) 2111–2118.
[10] M. Elhadidy, S. Shaahid, Parametric study of hybrid (wind + solar + diesel) power generating systems, Renewable Energy 21 (2) (2000) 129–139.
[11] S. Rehman, L.M. Al-Hadhrami, Study of a solar PV–diesel–battery hybrid power system for a remotely located population near Rafha, Saudi Arabia, Energy 35 (12) (2010) 4986–4995.
[12] D. Xu, L. Kang, L. Chang, and B. Cao, ''Optimal sizing of standalone hybrid wind/PV power systems using genetic algorithms,'' in Canadian Conference on Electrical and Computer Engineering, 2005., 2005, pp. 1722-1725: IEEE.
[13] S. Diaf, D. Diaf, M. Belhamel, M. Haddadi, A. Louche, A methodology for optimal sizing of autonomous hybrid PV/wind system, Energy policy 35 (11) (2007) 5708–5718.
[14] M. o. E. a. E. i. Egypt. (2020, 15.01.2020). New and Renewable Energy Authority. Available: http://www.nrea.gov.eg/Technology/ProjectLocation.
[15] H.M. Sultan, O.N. Kuznetsov, A.A.Z. Diab, Site selection of large-scale grid-connected solar PV system in Egypt, in: 2018 IEEE Conference of Russian Young Researchers in Electrical and Electronic Engineering (EIConRus), IEEE, 2018, pp. 813–818.
[16] M. o. E. a. Energy, ''2016/2017 Technical report of the Egyptian Holding Company,'' 2017.
[17] Y. Sawle, S. Gupta, A.K. Bohre, Optimal sizing of standalone PV/Wind/Biomass hybrid energy system using GA and PSO optimization technique, Energy Procedia 117 (2017) 690–698.
[18] R. Dufo-López et al, Multi-objective optimization minimizing cost and life cycle emissions of stand-alone PV–wind–diesel systems with batteries storage, Appl. Energy 88 (11) (2011) 4033–4041.
[19] S. Abedi, H. G. Ahangar, M. Nick, and S. H. Hosseinian, ''Economic and reliable design of a hybrid PV-wind-fuel cell energy system using differential evolutionary algorithm,'' in 2011 19th Iranian Conference on Electrical Engineering, 2011, pp. 1-6: IEEE.
[20] S. Zhang, P. Huang, Y. Sun, A multi-criterion renewable energy system design optimization for net zero energy buildings under uncertainties, Energy 94 (2016) 654–665.
[21] T. Ma, H. Yang, L. Lu, J. Peng, Optimal design of an autonomous solar–wind-pumped storage power supply system, Appl. Energy 160 (2015) 728–736.
[22] M.A. Mohamed, A.M. Eltamaly, A.I. Alolah, A. Hatata, A novel framework-based cuckoo search algorithm for sizing and optimization of grid-independent hybrid renewable energy systems, Int. J. Green Energy 16 (1) (2019) 86–100.
[23] M.A. Ramli, H. Bouchekara, A.S. Alghamdi, Optimal sizing of PV/wind/diesel hybrid microgrid system using multi-objective self-adaptive differential evolution algorithm, Renewable Energy 121 (2018) 400–411.
[24] W. Dong, Y. Li, J. Xiang, Optimal sizing of a stand-alone hybrid power system based on battery/hydrogen with an improved ant colony optimization, Energies 9 (10) (2016) 785.
[25] S.A. Nowdeh, M. Hajibeigy, Economic designing of pv/fc/wind hybrid system considering components availability, Int. J. Modern Education Comput. Sci. 5 (7) (2013) 69.
[26] A.K. Kaviani, G. Riahy, S.M. Kouhsari, Optimal design of a reliable hydrogen-based stand-alone wind/PV generation system, in: 2008 11th International Conference on Optimization of Electrical and Electronic Equipment, IEEE, 2008, pp. 413–418.
[27] M. Ashari, H. Suryoatmojo, I. Robandi, T. Hiyama, Optimal design of hydrogen based stand-alone wind/microhydro system using genetic algorithm, ICSIIT 2010 (2010) 71.
[28] S. Upadhyay, M. Sharma, Selection of a suitable energy management strategy for a hybrid energy system in a remote rural area of India, Energy 94 (2016) 352–366.
[29] A. Kamjoo, A. Maheri, A.M. Dizqah, G.A. Putrus, Multi-objective design under uncertainties of hybrid renewable energy system using NSGA-II and chance constrained programming, Int. J. Electr. Power Energy Syst. 74 (2016) 187–194.
[30] S. Mohammadi, S. Soleymani, B. Mozafari, Scenario-based stochastic operation management of microgrid including wind, photovoltaic, micro-turbine, fuel cell and energy storage devices, Int. J. Electr. Power Energy Syst. 54 (2014) 525–535.
[31] A. Maleki, Design and optimization of autonomous solar-wind-reverse osmosis desalination systems coupling battery and hydrogen energy storage by an improved bee algorithm, Desalination 435 (2018) 221–234.
[32] A. Maleki, Optimal operation of a grid-connected fuel cell based combined heat and power systems using particle swarm optimisation for residential sector, Int. J. Ambient Energy (2019) 1–8.
[33] W. Zhang, A. Maleki, M.A. Rosen, A heuristic-based approach for optimizing a small independent solar and wind hybrid power scheme incorporating load forecasting, J. Cleaner Prod. 241 (2019) 117920.
[34] A. Maleki, M.A. Rosen, Design of a cost-effective on-grid hybrid wind–hydrogen based CHP system using a modified heuristic approach, Int. J. Hydrogen Energy 42 (25) (2017) 15973–15989.
[35] W. Zhang, A. Maleki, M.A. Rosen, J. Liu, Sizing a stand-alone solar-wind-hydrogen energy system using weather forecasting and a hybrid search optimization algorithm, Energy Convers. Manage. 180 (2019) 609–621.
[36] A. Hatata, G. Osman, M. Aladl, An optimization method for sizing a solar/wind/battery hybrid power system based on the artificial immune system, Sustainable Energy Technol. Assess. 27 (2018) 83–93.
[37] A. A. Z. Diab, S. I. El-ajmi, H. M. Sultan, and Y. B. Hassan, ''Modified farmland fertility optimization algorithm for optimal design of a grid-connected hybrid renewable energy system with fuel cell storage: case study of Ataka, Egypt,'' International Journal of Advanced Computer Science and Applications, vol. 10, no. 8, 2019.
[38] R. Kaur, V. Krishnasamy, N.K. Kandasamy, S. Kumar, Discrete multiobjective grey wolf algorithm based optimal sizing and sensitivity analysis of PV-wind-battery system for rural telecom towers, IEEE Syst. J. 14 (1) (2019) 729–737.
[39] A. Fathy, A reliable methodology based on mine blast optimization algorithm for optimal sizing of hybrid PV-wind-FC system for remote area in Egypt, Renewable Energy 95 (2016) 367–380.
[40] K. Gholami, E. Dehnavi, A modified particle swarm optimization algorithm for scheduling renewable generation in a micro-grid under load uncertainty, Appl. Soft Comput. 78 (2019) 496–514.

[41] M. Samy, S. Barakat, H. Ramadan, Techno-economic analysis for rustic electrification in Egypt using multi-source renewable energy based on PV/wind/FC, Int. J. Hydrogen Energy 45 (20) (2019) 11471–11483.
[42] S.A. Shezan, Optimization and assessment of an off-grid photovoltaic–diesel–battery hybrid sustainable energy system for remote residential applications, Environ. Prog. Sustainable Energy 38 (6) (2019) e13340.
[43] M. Nurunnabi, N.K. Roy, E. Hossain, H.R. Pota, Size Optimization and Sensitivity Analysis of Hybrid Wind/PV Micro-Grids-A Case Study for Bangladesh, IEEE Access 7 (2019) 150120–150140.
[44] O.N. Kuznetsov, H.M. Sultan, R.I. Aljendy, A.A.Z. Diab, Economic Feasibility Analysis of PV/Wind/Diesel/Battery Isolated Microgrid for Rural Electrification in South Egypt, in: 2019 IEEE Conference of Russian Young Researchers in Electrical and Electronic Engineering (EIConRus), IEEE, 2019, pp. 1001–1006.
[45] S. Shezan, A. Al-Mamoon, H. Ping, Performance investigation of an advanced hybrid renewable energy system in Indonesia, Environ. Prog. Sustainable Energy 37 (4) (2018) 1424–1432.
[46] B. Yan, P.B. Luh, G. Warner, P. Zhang, Operation and design optimization of microgrids with renewables, IEEE Trans. Autom. Sci. Eng. 14 (2) (2017) 573–585.
[47] A.K. Kaviani, G. Riahy, S.M. Kouhsari, Optimal design of a reliable hydrogen-based stand-alone wind/PV generating system, considering component outages, Renewable Energy 34 (11) (2009) 2380–2390.
[48] T. Ma, H. Yang, L. Lu, Feasibility study and economic analysis of pumped hydro storage and battery storage for a renewable energy powered island, Energy Convers. Manage. 79 (2014) 387–397.
[49] T. Ma, H. Yang, L. Lu, J. Peng, Technical feasibility study on a standalone hybrid solar-wind system with pumped hydro storage for a remote island in Hong Kong, Renewable Energy 69 (2014) 7–15.
[50] H.M. Sultan, A.A.Z. Diab, N.K. Oleg, S.Z. Irina, Design and evaluation of PV-wind hybrid system with hydroelectric pumped storage on the National Power System of Egypt, Global Energy Interconnection 1 (3) (2018) 301–311.
[51] I. E. Commission, ‘‘Wind turbines: Part 3: Design requirements for offshore wind turbines; 2009,’’ Google Scholar, 2009.
[52] R.S. Garcia, D. Weisser, A wind–diesel system with hydrogen storage: Joint optimisation of design and dispatch, Renewable Energy 31 (14) (2006) 2296–2320.
[53] A. Mills, S. Al-Hallaj, Simulation of hydrogen-based hybrid systems using Hybrid2, Int. J. Hydrogen Energy 29 (10) (2004) 991–999.
[54] M. Khan, M. Iqbal, Pre-feasibility study of stand-alone hybrid energy systems for applications in Newfoundland, Renewable Energy 30 (6) (2005) 835–854.
[55] M. El-Sharkh, M. Tanrioven, A. Rahman, M. Alam, Cost related sensitivity analysis for optimal operation of a grid-parallel PEM fuel cell power plant, J. Power Sources 161 (2) (2006) 1198–1207.
[56] K. Strunz, E.K. Brock, Stochastic energy source access management: infrastructure-integrative modular plant for sustainable hydrogen-electric co-generation, Int. J. Hydrogen Energy 31 (9) (2006) 1129–1141.
[57] L. Xu, X. Ruan, C. Mao, B. Zhang, Y. Luo, An improved optimal sizing method for wind-solar-battery hybrid power system, IEEE Trans. Sustainable Energy 4 (3) (2013) 774–785.
[58] A. A. Z. Diab, S. I. EL-Ajmi, H. M. Sultan, and Y. B. Hassan, ‘‘Modified Farmland Fertility Optimization Algorithm for Optimal Design of a Grid-connected Hybrid Renewable Energy System with Fuel Cell Storage: Case Study of Ataka, Egypt.’’
[59] W. Zhao, L. Wang, Z. Zhang, Artificial ecosystem-based optimization: a novel nature-inspired meta-heuristic algorithm, Neural Comput. Appl. (2019) 1–43.
[60] A.S. Menesy, H.M. Sultan, A. Korashy, F.A. Banakhr, M.G. Ashmawy, S. Kamel, Effective parameter extraction of different polymer electrolyte membrane fuel cell stack models using a modified artificial ecosystem optimization algorithm, IEEE Access 8 (2020) 31892–31909.
[61] J. Kennedy and R. Eberhart, ‘‘Particle swarm optimization,’’ in Proceedings of ICNN’95-International Conference on Neural Networks, 1995, vol. 4, pp. 1942-1948: IEEE.
[62] S. Mirjalili, A.H. Gandomi, S.Z. Mirjalili, S. Saremi, H. Faris, S. M. Mirjalili, Salp swarm algorithm: a bio-inspired optimizer for engineering design problems, Adv. Eng. Softw. 114 (2017) 163–191.
[63] S. Mirjalili, S.M. Mirjalili, A. Lewis, Grey wolf optimizer, Adv. Eng. Softw. 69 (2014) 46–61.
[64] B.S. Yıldız, A.R. Yıldız, Comparison of grey wolf, whale, water cycle, ant lion and sine-cosine algorithms for the optimization of a vehicle engine connecting rod, Materials Testing 60 (3) (2018) 311–315.